Exact Diamond Balls, Antipodal Widths, and Query-Uniform Compression Bounds for Quantum Channels

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Abstract

Quantum channels with fixed finite-dimensional input and output form a compact convex body. We ask how many continuous latent parameters are needed to encode an arbitrary channel and reconstruct it with uniformly small diamond-norm error, when the encoder is continuous and the decoder is unrestricted. This is a continuous analogue of nonlinear width, distinct from tomography, which counts channel uses, and from programmable processors, which fix a decoding device. For a compact source the optimal error equals the minimal worst-case radius of the encoder's fibers. Lower bounds follow from antipodal pairs in the source and from latent collision topology via Borsuk–Ulam and the deleted-product coindex. Even input-independent output preparation forces error one below the output-state dimension, with a Clifford analogue inside the unitary channels. We determine the largest diamond ball contained in the channel body: it is centred at the completely depolarizing channel, is optimal among all centres, and, together with a full-direction section, certifies the intermediate lower envelope; the same channel is the optimal constant decoder, with exact covering radius, so the two envelopes meet at the exact width at zero latent dimension. Constructive upper bounds from convex projection and balanced pinching bracket the intermediate widths, and meet the lower envelope in the degenerate collapse case and from the affine dimension on. Every finite-query adaptive network induces a continuous final code containing all side information, so the same bottlenecks hold uniformly in the number of queries, for uniformly successful postselection, and for sampled transcripts; a one-query informationally complete measurement achieves the injective threshold, and repetition gives an $O(N^{-1/2})$ expected-error bound.

Keywords: quantum channel compression; diamond norm; antipodal width; exact in-radius; Borsuk–Ulam theorem; quantum combs; nonlinear approximation.

1 Introduction

For fixed finite-dimensional input and output systems, quantum channels form a compact convex body. The question is whether this body admits a continuous low-dimensional encoding with uniformly accurate reconstruction in diamond norm.

The problem differs from tomography and programming. Tomography asks how many uses of an unknown channel are needed to produce an estimator. A programmable processor uses a fixed physical device to transform a program state into an implemented channel [1]. Here continuity is imposed on the encoder, while the decoder is initially unrestricted. The problem is therefore an encoder–decoder variant of continuous nonlinear approximation [2]. Stable manifold methods and Lipschitz widths provide related regularity-sensitive benchmarks [3, 4], while topological limitations of autoencoding are studied in [5, 6]. Variational quantum-circuit autoencoders give an architecture-specific channel-compression model [7]; the bounds here instead range over every continuous encoder with the stated latent type.

Query complexity and retained representation dimension are different resources: more channel uses may improve tomography, while a fixed low-dimensional continuous code can remain topologically unable to represent the full channel body. Related distinctions between query and memory resources occur in quantum process learning [8]. Encoder continuity is an explicit representational hypothesis, not a claim about every possible synthesis map. Deterministic finite-query networks provide a broad physical class for which this hypothesis follows from a proved Lipschitz estimate. Robustness to perturbations of the stored code is a separate decoder-regularity question.

The abstract theory separates three objects. First, unrestricted decoding allows an exact fiber-radius characterization of reconstruction width. Second, an antipodal separation profile measures the metric size of spherical sections in the source. Third, the $\mathbb{Z}_2$-coindex of the deleted product of a latent space determines which spherical maps must have an antipodal collision. Euclidean Borsuk–Ulam is recovered as the basic case.

The channel application has two geometric scales. The full output-state space, embedded by replacement channels, gives perfect antipodal separation through index $d_B^2 - 2$. Across every remaining subcritical affine dimension, the exact depolarizing-centred ball gives the lower bound $2/[d_B \min\{d_A, d_B\}]$. Conversely, affine projection gives a universal upper envelope near the affine threshold, while physical input/output pinching retains selected coherent sectors and gives stronger structured upper bounds at lower dimensions. Structured block joins are retained in the appendix, and anticommuting Clifford generators give a perfectly separated sphere inside the unitary channels for every square system of dimension at least two.

A deterministic finite adaptive network, equivalently a finite quantum comb [9] and quantum strategy [10], produces a continuous final code. Every quantum or classical register available to reconstruction is part of that code. The result applies to every finite query count and to uniformly successful postselection on the channel domain under consideration, but not to conditioning on events whose probability can vanish. The unrestricted state-space decoder is distinguished from a fixed physical programmable processor.

The contributions are: (i) an exact fiber-radius characterization of encoder-continuous width with unrestricted decoder; (ii) a source–latent collision theorem via antipodal profile and deleted-product coindex, recovering Euclidean Borsuk–Ulam as a special case; (iii) a maximal metric lower bound 1 through index $L_{A,B}$ from the replacement family, the observable sphere, and their join, a full-direction section certifying $1/d_A$ through the affine dimension, and a Clifford unitary analogue; (iv) the exact all-dimensional depolarizing-centred in-radius — optimal among all centres — and the exact Chebyshev covering radius, with variational and spectral certificates; (v) constructive upper envelopes from convex projection and balanced input/output pinching, bracketing the widths below the affine threshold; (vi) query-uniform final-code bottlenecks for deterministic finite-query combs, uniformly successful postselection, and finite sampled transcripts, together with a one-query injective encoding and $O(N^{-1/2})$ finite-sample upper bound. The exact positive widths below the affine dimension remain open.

Section 2 develops fiber radii, antipodal profiles, and latent-space collisions. Section 3 then fixes the channel and norm conventions used in the applications. Section 4 studies relative diamond balls and centred in-radii. Section 5 develops the replacement, observable, full-direction, and Clifford sections and the resulting channel lower and constructive upper envelopes. Section 6 treats abstract state alphabets and physical programs. Section 7 gives the adaptive-network consequences, and Section 8 records scope and open questions. The appendix collects the longer proofs of the radius and covering results, fixed-marginal twirling, programming packing, structured partitions, and the Clifford construction. Ref. [11] extends the ball and compression theorems to finite-outcome quantum instruments and records, in an appendix, the no-programming background including the mixed-program and unequal-output cases; conversely, the channel-level theorems of the present paper are recovered from the instrument results by taking $n = 1$.

Throughout, all systems are finite-dimensional and all latent dimensions are real dimensions. Unless stated otherwise, the encoder is continuous and the decoder has no regularity requirement. The complete retained code is defined as the density operator, classical probability law, sampled symbol, or classical–quantum state containing all registers, outcomes, and side information available at reconstruction time before decoding; every classical transcript, quantum register, reference system, or side register available to the decoder is part of the code. An averaged (deterministic) code state is distinguished from a single-run sampled transcript. The diamond norm uses a reference system of input dimension d_A.

2 Source–latent obstructions

Let K be a compact subset of a metric space (X, d). The decoder target X may equal K or may be a larger ambient metric space.

2.1 Fiber radii and antipodal profiles

Definition 2.1 (Encoder-continuous reconstruction width)**.** For a topological latent space Y, define

$$\delta^{\mathrm{c}}(K; Y, X) := \inf_{\substack{f:K\to Y \text{ continuous} \\ g:Y\to X}} \sup_{x\in K} d(g(f(x)), x).$$

For $Y = \mathbb{R}^r$, write $\delta^{\mathrm{c}}_r(K; X)$, with $\mathbb{R}^0$ a singleton. No regularity is imposed on g. If no continuous $f : K \to Y$ exists (e.g., K nonempty and Y empty), define the infimum over an empty set to be $+\infty$.

This is a variant of continuous nonlinear width in which only the encoder is required to be continuous. We refer to this quantity as the compression width.

Definition 2.2 (Fiber radii and fiber diameter)**.** For nonempty $F \subseteq K$, set

$$\mathrm{rad}_X(F) = \inf_{z\in X} \sup_{x\in F} d(z, x).$$

For a topological latent space Y, define

$$\nu(K; Y, X) = \inf_{f\in C(K,Y)} \sup_{y\in f(K)} \mathrm{rad}_X(f^{-1}(y)).$$

For $Y = \mathbb{R}^r$, write $\nu_r(K; X)$. Also define

$$\mu_r(K) = \frac{1}{2} \inf_{f\in C(K,\mathbb{R}^r)} \sup_{y\in f(K)} \operatorname{diam} f^{-1}(y).$$

Theorem 2.1 (Exact fiber-radius characterization)**.** *For every topological latent space Y,*

$$\delta^{\mathrm{c}}(K; Y, X) = \nu(K; Y, X).$$

In particular, for every $r \in \mathbb{N}_0$,

$$\delta^{\mathrm{c}}_r(K; X) = \nu_r(K; X), \qquad \mu_r(K) \le \delta^{\mathrm{c}}_r(K; X) \le 2\mu_r(K).$$

Proof. Fix $f \in C(K, Y)$. A decoder value $g(y)$ affects only the fiber $f^{-1}(y)$, so its least possible worst error on that fiber is $\mathrm{rad}_X(f^{-1}(y))$. Thus every decoder has error at least the supremum of the fiber radii. Conversely, for every $\varepsilon > 0$ and every $y \in f(K)$, choose $z_y \in X$ such that

$$\sup_{x\in f^{-1}(y)} d(z_y, x) \le \mathrm{rad}_X(f^{-1}(y)) + \varepsilon.$$

Define $g(y) = z_y$ on $f(K)$ and define it arbitrarily on $Y \setminus f(K)$. No compatibility between these choices is required because the decoder has no regularity assumption. The resulting error is at most the supremum of the fiber radii plus ε. Letting $\varepsilon \downarrow 0$ and then taking the infimum over f proves the identity. If X is compact, the radius function $z \mapsto \sup_{x\in F} d(z, x)$ is 1-Lipschitz, hence continuous, since $|\sup_{x\in F} d(z,x) - \sup_{x\in F} d(z',x)| \le d(z,z')$, and therefore attains its infimum on X for every nonempty $F \subseteq K$. The same conclusion holds when X is proper: F is bounded because $F \subseteq K$, and the centre search may be restricted to a sufficiently large compact ball (closed balls are compact in a proper metric space). In either case the ε-selection is unnecessary.

For every nonempty $F \subseteq K \subseteq X$,

$$\frac{1}{2}\operatorname{diam} F \le \operatorname{rad}_X(F) \le \operatorname{diam} F.$$

The first inequality is the triangle inequality. For the second, choose a point of F as centre. Taking the relevant suprema and infima gives the factor-two comparison. $\square$

Remark 2.2 (Fiber picture). The encoder f partitions K into fibers $f^{-1}(y)$. For fixed f, the decoder value $g(y)$ can be optimized independently for each fiber because no regularity is imposed on g; the optimal worst-case error on the fiber is its radius rad_X. The width $\nu(K;Y,X)$ is therefore the minimal possible maximal fiber radius achievable by a continuous f.

Definition 2.3 (Antipodal separation profile)**.** For $r \in \mathbb{N}_0$, define

$$\alpha_r(K) = \frac{1}{2}\sup_{h\in C(S^r,K)} \min_{u\in S^r} d(h(u), h(-u)).$$

The minimum exists by compactness of S^r. The quantity $\alpha_r(K)$ is henceforth abbreviated to the antipodal profile.

Proposition 2.3 (Profile properties)**.** *Let K be a compact metric space. In items involving a subspace $F \subseteq K$, assume that F is compact and carries the restricted metric. Then the following hold.*

1. $\alpha_{r+1}(K) \le \alpha_r(K)$.
2. *If $F \subseteq K$, then $\alpha_r(F) \le \alpha_r(K)$.*
3. *If F is compact, $T : F \to K$ is continuous and $d(Tx,Ty) \ge c\, d(x,y)$ with $c \ge 0$, then $\alpha_r(K) \ge c\alpha_r(F)$.*
4. $0 \le \alpha_r(K) \le \operatorname{diam}(K)/2$.
5. *If two metrics d_0, d_1 on K induce the same topology (as any two norm-induced metrics on a finite-dimensional ambient space do) and satisfy $|d_0(x,y) - d_1(x,y)| \le \varepsilon$ uniformly on K^2, then $|\alpha_r(K,d_0) - \alpha_r(K,d_1)| \le \varepsilon/2$. Uniform closeness alone does not suffice: the perturbation $d_1(x,y) = d_0(x,y) + \varepsilon$ for $x \ne y$ is a uniformly ε-close metric inducing the discrete topology, for which discontinuous antipodal selectors enter the supremum.*

Proof. Restricting a map on S^{r+1} to an equatorial S^r proves the first claim. The second follows because maps into F are maps into K. For the third, compose each spherical map with T and use the distance inequality. For every spherical map, each antipodal separation lies between zero and $\operatorname{diam}(K)$, which proves the fourth claim after taking the minimum, supremum, and factor one half. For the fifth, the common topology makes $C(S^r,K)$ the same class of maps for both metrics; for every fixed h in this common class,

$$\left|\min_u d_0(h(u),h(-u)) - \min_u d_1(h(u),h(-u))\right| \le \varepsilon.$$

Taking suprema over the common class preserves this estimate, and the factor one half gives the stated bound. $\square$

Theorem 2.4 (Antipodal and fiber lower bounds)**.** *For every r,*

$$\alpha_r(K) \leq \mu_r(K) \leq \delta_r^{\mathrm{c}}(K; X) \leq 2\mu_r(K).$$

In particular,

$$\delta_r^{\mathrm{c}}(K; X) \geq \alpha_r(K).$$

Proof. Fix $f \in C(K, \mathbb{R}^r)$ and $h \in C(S^r, K)$. Borsuk–Ulam [12] gives u with $f(h(u)) = f(h(-u))$. Hence one fiber has diameter at least

$$d(h(u), h(-u)) \geq \min_v d(h(v), h(-v)).$$

Take the supremum over h, then the infimum over f, and divide by two to obtain $\alpha_r(K) \leq \mu_r(K)$. The remaining inequalities are Theorem 2.1. $\square$

2.2 Intrinsic topology of the latent space

The deleted product supplies an intrinsic collision criterion for the latent space, independent of a chosen Euclidean embedding.

Definition 2.4 (Deleted-product coindex)**.** For a Hausdorff space Y, let

$$F_2(Y) = \{(y_0, y_1) \in Y^2 : y_0 \neq y_1\},$$

with involution $(y_0, y_1) \mapsto (y_1, y_0)$. Define

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) = \sup\{k \in \mathbb{N}_0 : \text{there exists a continuous equivariant map } S^k \to F_2(Y)\},$$

where S^k has the antipodal action. If $F_2(Y) = \varnothing$, the value is defined to be -1; otherwise the displayed supremum is used and may be $+\infty$. This is the standard sphere-map coindex convention; see, for example, [12]. When the acting group is clear, we occasionally suppress the subscript $\mathbb{Z}_2$.

Remark 2.5 (Configuration-space intuition). $F_2(Y)$ is the ordered configuration space of two distinct latent codes. An equivariant map $S^r \to F_2(Y)$ assigns to each u a distinct pair of codes for u and $-u$ continuously and respecting antipodal symmetry. If such a map does not exist, every continuous $f \circ h : S^r \to Y$ must identify some antipodal pair, forcing a collision. The coindex is the largest r for which collision can be avoided.

Theorem 2.6 (Intrinsic latent-space obstruction)**.** *Let Y be Hausdorff. If*

$$r > \operatorname{coind}_{\mathbb{Z}_2} F_2(Y),$$

then every continuous $f : K \to Y$ and every decoder $g : Y \to X$ satisfy

$$\sup_{x \in K} d(g(f(x)), x) \geq \alpha_r(K).$$

Proof. Fix $h : S^r \to K$. If $f(h(u)) \neq f(h(-u))$ for every u, then

$$u \longmapsto (f(h(u)), f(h(-u)))$$

is an equivariant map from S^r to $F_2(Y)$, contradicting the coindex hypothesis. Thus an antipodal pair has one code. The triangle inequality forces one reconstruction error to be at least half the pair separation. Take the supremum over h. An empty latent space admits no encoder from the nonempty source; by the convention in Definition 2.1 the width is then $+\infty$, so no separate case arises. $\square$

Proposition 2.7 (Euclidean recovery)**.** *If Y admits a continuous injection into $\mathbb{R}^s$, then*

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(Y) \leq s-1.$$

Consequently every continuous encoder through Y has error at least $\alpha_s(K)$. More generally, if a compact metric space K admits a continuous injection into $\mathbb{R}^s$, then $\alpha_r(K)=0$ for every $r \geq s$. If Y is a convex body with nonempty interior in $\mathbb{R}^s$, then equality holds in the coindex bound.

Proof. If $s=0$, injectivity into $\mathbb{R}^0$ forces Y to have at most one point. Thus $F_2(Y)=\varnothing$ and its coindex is $-1=s-1$. Likewise, any compact K that injects into $\mathbb{R}^0$ is a singleton, so $\alpha_r(K)=0$ for every $r \geq 0$.

Assume $s \geq 1$ and let $\iota : Y \to \mathbb{R}^s$ be injective. The normalised difference map

$$(y_0, y_1) \longmapsto \frac{\iota(y_0)-\iota(y_1)}{\|\iota(y_0)-\iota(y_1)\|_2}$$

is equivariant from $F_2(Y)$ to S^{s-1}. An equivariant map $S^k \to S^{s-1}$ cannot exist for $k \geq s$ by Borsuk–Ulam. Hence the coindex is at most $s-1$, and Theorem 2.6 applies with $r=s$.

If K admits a continuous injection $\kappa : K \to \mathbb{R}^s$ and $r \geq s$, then, for every $h : S^r \to K$, the generalised Borsuk–Ulam theorem (equivalently the nonexistence of an equivariant map $S^r \to S^{s-1}$ for $r \geq s$) applied to $\kappa \circ h$ gives a point u with $\kappa(h(u)) = \kappa(h(-u))$. Injectivity gives $h(u)=h(-u)$, so every such map has minimum antipodal separation zero and $\alpha_r(K)=0$.

For a full-dimensional convex body with $s \geq 1$, choose an interior point y_0 and $\varepsilon > 0$ such that $y_0 \pm \varepsilon u$ remain in the body for all $u \in S^{s-1}$. The map $u \mapsto (y_0+\varepsilon u, y_0-\varepsilon u)$ is equivariant into the deleted product, proving the reverse inequality. For $s=0$, such a body is a singleton and equality was already shown. □

2.3 Convex and stable certificates

The preceding results reduce reconstruction lower bounds to source separation and latent collisions. For convex sources, norm balls and their lower-dimensional sections provide explicit lower certificates, while affine projection and fiber centres provide a general upper construction. When the decoder is Lipschitz rather than unrestricted, the residual separation of the reachable latent image yields a quantitative stable form.

Theorem 2.8 (Contained-ball profile)**.** *Let K be a compact subset of a D-dimensional normed affine space, with its induced metric, and suppose $\overline{B}(x_0,\rho) \subseteq K$. Then*

$$\alpha_r(K) \geq \rho, \qquad 0 \leq r < D.$$

Proof. Choose an $(r+1)$-dimensional subspace W of the direction space, equip it with an auxiliary Euclidean norm, and let $S_E^r \subset W$ be its unit sphere; write $\|\cdot\|_{\mathrm{amb}}$ for the ambient norm. The radial map

$$h(u) = x_0 + \rho \frac{u}{\|u\|_{\mathrm{amb}}}$$

is continuous and satisfies $\|h(u)-h(-u)\|_{\mathrm{amb}} = 2\rho$. This map is admissible in Definition 2.3, so $\alpha_r(K) \geq \rho$. □

Definition 2.5 (Sectional in-radius)**.** Let K lie in a finite-dimensional normed affine space with direction space V. For $0 \leq r < \dim V$, define

$$R_r(K) = \sup_{\substack{x_0 \in K,\ W \subseteq V \\ \dim W = r+1}} \sup\{\rho \geq 0 : x_0 + \{w \in W : \|w\| \leq \rho\} \subseteq K\}.$$

Proposition 2.9 (Sectional-radius certificate)**.** *For every compact K in a finite-dimensional normed affine space,*

$$\alpha_r(K) \geq R_r(K).$$

Thus sectional convex geometry supplies profile lower bounds, while more general antipodal constructions, such as Clifford and join sections, need not arise as boundaries of norm balls.

Proof. For every admissible centre, subspace, and radius in Definition 2.5, apply the radial construction from Theorem 2.8 inside W. It produces an S^r with antipodal separation 2ρ. Taking the two suprema proves the claim. □

Proposition 2.10 (Convex-projection upper bound)**.** *Let K be a compact convex subset of a D-dimensional normed affine space, with diameter* $\operatorname{diam} K = \sup_{x,y\in K} \|x - y\|$*. Then, for $0 \leq r \leq D$,*

$$\delta_r^{\mathrm{c}}(K; K) \leq \frac{D-r}{D-r+1} \operatorname{diam} K.$$

Proof. For $r = D$, affine coordinates give exact reconstruction. Assume $0 \leq r < D$, put $k = D - r$, choose affine coordinates on $\operatorname{aff} K$, and project them onto their first r components. This gives a continuous encoder $f : K \to \mathbb{R}^r$. Every nonempty fiber $F_y = f^{-1}(y)$ is compact and convex, has diameter at most $\operatorname{diam} K$, and has affine dimension at most k. Write $d_y = \operatorname{diam} F_y$.

Fix such a fiber and set $R_y = k\, d_y/(k + 1)$. In the affine hull of F_y, consider the family

$$\mathcal{F}_y = \{F_y\} \cup \{\overline{B}(x, R_y) : x \in F_y\}.$$

Work in the normed affine space $\operatorname{aff}(F_y)$, of dimension at most k, and replace each ambient ball $\overline{B}(x, R_y)$ by its intersection with $\operatorname{aff}(F_y)$, which remains convex and becomes compact after intersecting with F_y. Every subfamily of at most $k + 1$ members has nonempty intersection. Indeed, if it consists of $m \leq k + 1$ balls centred at $x_1, \dots, x_m \in F_y$, their barycentre $c = m^{-1} \sum_j x_j$ lies in F_y and satisfies

$$\|c - x_i\| \leq \frac{1}{m} \sum_{j\neq i} \|x_j - x_i\| \leq \frac{m-1}{m}\, d_y \leq R_y.$$

If the subfamily also contains F_y, then it contains at most k balls, and the same barycentre belongs to F_y and satisfies the same estimate. The case of no balls is immediate.

Helly's theorem [13] for convex sets in $\mathbb{R}^k$ now gives that every finite subfamily of $\mathcal{F}_y$ has nonempty intersection. Since we may intersect each member with the compact set F_y, the family $\{F_y\} \cup \{F_y \cap \overline{B}(x, R_y) : x \in F_y\}$ consists of compact sets with the finite-intersection property. Hence there exists

$$c_y \in F_y \cap \bigcap_{x\in F_y} \overline{B}(x, R_y).$$

Define the decoder by $g(y) = c_y$ on $f(K)$ and arbitrarily elsewhere. Then

$$\|g(f(x)) - x\| \leq \frac{k}{k+1} d_{f(x)} \leq \frac{D-r}{D-r+1} \operatorname{diam} K$$

for every $x \in K$, which proves the claim. □

Proposition 2.11 (Exact Euclidean-coordinate threshold)**.** *Let K be compact, let its affine hull have dimension D, and equip it with the metric induced by a norm on that affine hull. Then $\delta_r^{\mathrm{c}}(K; K) = 0$ for $r \geq D$, and the exact decoder may be chosen continuous. If K contains a positive relative ball—that is, a ball of positive radius $\rho > 0$ in the relative topology of its affine hull, i.e., some $\overline{B}(x_0, \rho) = \{x_0 + w : \|w\| \leq \rho\} \subseteq K$ with w in the direction space of* $\operatorname{aff} K$ *—then*

$$\delta_r^{\mathrm{c}}(K; K) = 0 \iff r \geq D.$$

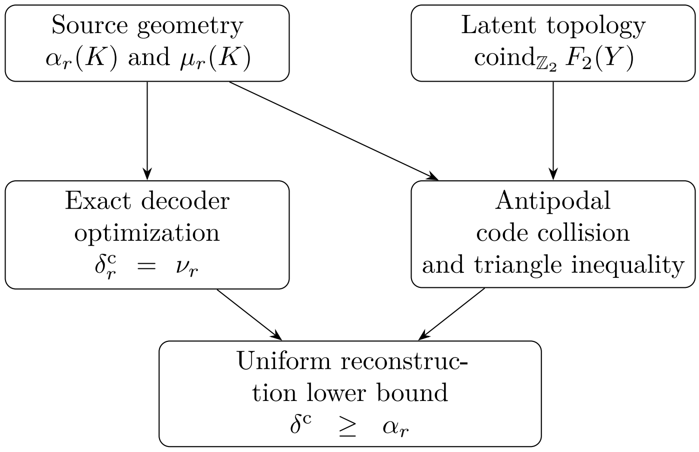


Figure 1: Logical structure of the source–latent obstruction. Fiber radii characterize unrestricted decoder optimization; antipodal source separation and latent collision topology provide computable lower bounds.

Proof. Affine coordinates c, padded by zeros, give a continuous injection into $\mathbb{R}^r$ for $r \geq D$. The image $c(K)$ is compact and convex; the Euclidean nearest-point projection Π_C onto a nonempty closed convex set is single-valued and 1-Lipschitz, hence continuous, and the decoder $g = c^{-1} \circ \Pi_C \circ$ (coordinate projection) is exact and continuous on the encoder image, extended arbitrarily elsewhere, which gives width zero. If K contains $\overline{B}(x_0, \rho)$ with $\rho > 0$, then for every $r < D$, Theorems 2.4 and 2.8 give $\delta_r^{\mathrm{c}}(K; K) \geq \rho > 0$. Thus the infimum cannot vanish below D, proving the converse implication without assuming that an infimizing encoder–decoder pair exists. □

Theorem 2.12 (Stable source–latent obstruction)**.** *Let $f : K \to Y$ be continuous, where Y is a metric space, and let $g : Y \to X$ be L-Lipschitz. Then*

$$\sup_{x \in K} d(g(f(x)), x) \geq \alpha_r(K) - L\alpha_r(f(K)),$$

where $f(K)$ carries the metric induced from Y.

Proof. The set $f(K)$ is compact. Fix $h : S^r \to K$ and set $a_h = \min_u d(h(u), h(-u))$. The map $u \mapsto d_Y(f(h(u)), f(h(-u)))$ is continuous on the compact sphere, so it has a minimizer u_0. By the definition of the profile of $f(K)$,

$$d_Y(f(h(u_0)), f(h(-u_0))) \leq 2\alpha_r(f(K)).$$

If E is the worst reconstruction error, then by the definition of a_h, the triangle inequality, and the Lipschitz property,

$$\begin{aligned} a_h &\leq d(h(u_0), h(-u_0)) \\ &\leq d(h(u_0), g(f(h(u_0)))) + d(g(f(h(u_0))), g(f(h(-u_0)))) + d(g(f(h(-u_0))), h(-u_0)) \\ &\leq 2E + L\, d_Y(f(h(u_0)), f(h(-u_0))) \\ &\leq 2E + 2L\alpha_r(f(K)). \end{aligned}$$

Taking the supremum over h and dividing by two proves the claim. □

Figure 1 summarises the roles of fiber optimization, source separation, and latent collision topology.

3 Channel geometry and norm conventions

Let A and B be nonzero finite-dimensional Hilbert spaces, with $d_A = \dim A$ and $d_B = \dim B$. Write $\mathcal{L}(A)$ for the linear operators on A, and let $\mathbf{Chan}(A, B)$ denote the channels $\Phi : \mathcal{L}(A) \to \mathcal{L}(B)$. Let $A_0 \simeq A$ be a fixed reference copy and set

$$|\omega_A\rangle = \frac{1}{\sqrt{d_A}} \sum_{j=1}^{d_A} |j\rangle_{A_0} |j\rangle_A.$$

The normalised Choi operator of Φ is [14, 15]

$$\widehat{J}_\Phi = (\mathrm{id}_{A_0} \otimes \Phi)(|\omega_A\rangle\langle\omega_A|).$$

Equivalently, with column vectorization $|K\rangle\rangle = \sum_{a,b} K_{ba} |a\rangle_{A_0} |b\rangle_B$, the unnormalized Choi operator of a completely positive map with Kraus operators K_α is $J_\Phi = \sum_\alpha |K_\alpha\rangle\rangle\langle\langle K_\alpha|$. Then

$$\Phi \in \mathbf{Chan}(A, B) \quad \Longleftrightarrow \quad \widehat{J}_\Phi \geq 0, \qquad \mathrm{Tr}_B \widehat{J}_\Phi = \frac{I_{A_0}}{d_A}.$$

The affine direction space is

$$V_{A,B} = \{X \in \mathrm{Herm}(A_0 \otimes B) : \mathrm{Tr}_B X = 0\},$$

so

$$D_{A,B} := \dim_{\mathrm{aff}} \mathbf{Chan}(A, B) = d_A^2 (d_B^2 - 1). \tag{1}$$

If $d_B = 1$, the channel body is a singleton. Henceforth $d_B > 1$.

The completely depolarizing channel is

$$\Omega_{A,B}(X) = \mathrm{Tr}(X) \frac{I_B}{d_B}, \qquad \widehat{J}_{\Omega_{A,B}} = \frac{I_{A_0 \otimes B}}{d_A d_B}.$$

Its Choi operator is positive definite, so it is a relative interior point; convex-geometric analyses of related quantum bodies are carried out in [16]. For later use, the replacement channel associated with $\sigma \in \mathcal{S}(B)$ is

$$\mathcal{R}_\sigma(X) = \mathrm{Tr}(X)\sigma.$$

For a Hermiticity-preserving map $\Delta : \mathcal{L}(A) \to \mathcal{L}(B)$,

$$\|\Delta\|_\diamond = \sup_{\rho \in \mathcal{S}(A_0 \otimes A)} \|(\mathrm{id}_{A_0} \otimes \Delta)(\rho)\|_1.$$

A reference of dimension d_A suffices [17]. Operationally, for two equally likely channels Φ and Ψ, the optimal single-use discrimination error is $\frac{1}{2} - \frac{1}{4}\|\Phi - \Psi\|_\diamond$, allowing arbitrary entangled inputs and final measurements. Since the maximally entangled state is one admissible input,

$$\|\widehat{J}_\Delta\|_1 \leq \|\Delta\|_\diamond. \tag{2}$$

Lemma 3.1 (Trace-zero spectral bound)**.** *If $X = X^\dagger$ and $\mathrm{Tr}\, X = 0$, then*

$$\|X\|_\infty \leq \frac{1}{2}\|X\|_1.$$

Proof. Write $X = X_+ - X_-$ for its Jordan decomposition. Then $X_\pm \geq 0$, $X_+ X_- = 0$, and $\mathrm{Tr}\, X_+ = \mathrm{Tr}\, X_- = \|X\|_1/2$. For positive semidefinite P, $\|P\|_\infty \leq \mathrm{Tr}\, P = \|P\|_1$. Hence

$$\|X\|_\infty = \max\{\|X_+\|_\infty, \|X_-\|_\infty\} \leq \frac{1}{2}\|X\|_1.$$

□

Lemma 3.2 (Choi bounds in the present convention)**.** *Let* $J_\Delta = d_A \widehat{J}_\Delta$ *be the unnormalized Choi operator of a Hermiticity-preserving map. Then*

$$\frac{1}{d_A}\|J_\Delta\|_1 \le \|\Delta\|_\diamond \le \|\operatorname{Tr}_B |J_\Delta|\|_\infty.$$

Proof. The lower bound is (2). For the upper bound, write the Jordan decomposition $J_\Delta = J_+ - J_-$ with $J_\pm \ge 0$ of orthogonal support, and let $\Delta_\pm$ be the completely positive maps with Choi operators $J_\pm$ (in the unnormalized convention $J_\Delta = \sum_{i,j} |i\rangle\langle j| \otimes \Delta(|i\rangle\langle j|)$, with trace over the output factor; the same bound for Hermiticity-preserving maps is derived independently in [18, Corollary 2]). For every state ρ on a reference-and-input system, $(\mathrm{id}\otimes\Delta)(\rho) = (\mathrm{id}\otimes\Delta_+)(\rho) - (\mathrm{id}\otimes\Delta_-)(\rho)$ is a difference of positive operators, so its trace norm is at most the sum of their traces. Computing each trace through the adjoint gives $\operatorname{Tr}[(\mathrm{id}\otimes\Delta_\pm)(\rho)] = \operatorname{Tr}[\rho\,(I\otimes\operatorname{Tr}_B J_\pm)]$, so the sum equals $\operatorname{Tr}[\rho\,(I\otimes\operatorname{Tr}_B|J_\Delta|)]$, which is at most $\|\operatorname{Tr}_B |J_\Delta|\|_\infty$ because ρ is a state. Taking the supremum over ρ proves the upper bound. We shall also use the elementary estimate

$$\|\operatorname{Tr}_B |J_\Delta|\|_\infty \le \operatorname{Tr}(\operatorname{Tr}_B |J_\Delta|) = \|J_\Delta\|_1,$$

which follows because $\operatorname{Tr}_B |J_\Delta|$ is positive. □

4 Relative diamond balls and centred in-radii

The contained-ball profile theorem turns a channel-space in-radius into a compression lower bound. This section therefore determines the largest relative diamond ball about the completely depolarizing channel, first stating the exact value and then isolating its spectral and variational mechanisms.

A map Δ is trace-annihilating if $\operatorname{Tr}\Delta(X) = 0$ for all X, equivalently $\operatorname{Tr}_B \widehat{J}_\Delta = 0$. Define

$$\overline{B}^{\mathrm{rel}}_\diamond(\Phi_0,\rho) = \{\Phi_0 + \Delta : \Delta \text{ is Hermiticity-preserving and trace-annihilating, } \|\Delta\|_\diamond \le \rho\}.$$

Definition 4.1 (Centered relative diamond in-radius)**.** For $\Phi_0 \in \mathbf{Chan}(A,B)$, define

$$\rho_\diamond(\Phi_0) = \sup\{\rho \ge 0 : \overline{B}^{\mathrm{rel}}_\diamond(\Phi_0,\rho) \subseteq \mathbf{Chan}(A,B)\}.$$

Definition 4.2 (Variational parameter)**.** Define

$$\Gamma_{A,B} = \sup\left\{\lambda_{\max}(-\widehat{J}_\Delta) : \Delta \text{ is Hermiticity-preserving and trace-annihilating, } \|\Delta\|_\diamond \le 1\right\}.$$

4.1 Exact centred radius

Theorem 4.1 (Exact depolarizing-centred in-radius)**.** *Let*

$$s = \min\{d_A, d_B\}.$$

Then

$$\Gamma_{A,B} = \frac{s}{2d_A}, \qquad \rho_\diamond(\Omega_{A,B}) = \rho_{\mathrm{dep}}(A,B) := \frac{2}{d_B s}.$$

Equivalently,

$$\rho_\diamond(\Omega_{A,B}) = \begin{cases} 2/(d_A d_B), & d_A \le d_B, \\ 2/d_B^2, & d_A > d_B. \end{cases}$$

The proof is given in Appendix A.1.

The exact theorem separates two dimension regimes. When $d_A \le d_B$, an isometric channel on the full input space is extremal. When $d_A > d_B$, the extremal construction is supported on a d_B-dimensional input subspace, and the radius ceases to depend on d_A. Operationally, the relevant perturbation can be detected with a maximally entangled probe whose Schmidt rank is at most $s = \min\{d_A, d_B\}$. Extra input dimensions beyond the output capacity do not produce a direction with a smaller positivity radius: the witness already fits inside an output-sized input subspace. Thus s, rather than the full input dimension alone, is the effective entanglement rank controlling robustness of the completely noisy channel against arbitrary trace-preserving Hermitian perturbations.

The following spectral certificate gives a direct but generally nonoptimal contained ball; the variational formula then identifies the exact radial optimization.

4.2 Spectral and variational mechanisms

Proposition 4.2 (Centre-dependent spectral certificate)**.** *Let $\Phi_0 \in \mathbf{Chan}(A, B)$ have positive-definite normalised Choi operator and set $\lambda_* = \lambda_{\min}(\widehat{J}_{\Phi_0})$. Then*

$$\overline{B}_{\diamond}^{\mathrm{rel}}(\Phi_0, 2\lambda_*) \subseteq \mathbf{Chan}(A, B).$$

Among channel centres, the depolarizing channel uniquely maximises the certificate $2\lambda_{\min}(\widehat{J}_{\Phi_0})$.

Proof. For an admissible perturbation $X = \widehat{J}_\Delta$, Lemma 3.1 and (2) give $\|X\|_\infty \le \|\Delta\|_\diamond/2$. Thus $\|\Delta\|_\diamond \le 2\lambda_*$ implies $\widehat{J}_{\Phi_0} + X \ge 0$; the marginal constraint is unchanged.

Every normalised Choi operator has trace one on a space of dimension $d_A d_B$, so $\lambda_* \le 1/(d_A d_B)$. Equality forces every eigenvalue to equal $1/(d_A d_B)$, hence $\widehat{J}_{\Phi_0} = I/(d_A d_B)$ and $\Phi_0 = \Omega_{A,B}$. □

At the depolarizing centre, Proposition 4.2 gives the explicit certificate

$$\overline{B}_{\diamond}^{\mathrm{rel}}\left(\Omega_{A,B}, \frac{2}{d_A d_B}\right) \subseteq \mathbf{Chan}(A, B),$$

because $\widehat{J}_{\Omega_{A,B}} = I/(d_A d_B)$. This certificate equals the true centred radius when $d_A \le d_B$ and is strictly smaller when $d_A > d_B$.

Proposition 4.3 (Variational formula for the centred in-radius)**.** *The parameter $\Gamma_{A,B}$ satisfies*

$$\rho_\diamond(\Omega_{A,B}) = \frac{1}{d_A d_B\, \Gamma_{A,B}}.$$

Moreover, $0 < \Gamma_{A,B} \le 1/2$.

Proof. Let $N = d_A d_B$ and let $\mathcal{D}$ be the diamond unit sphere in the finite-dimensional real vector space of Hermiticity-preserving, trace-annihilating maps. The set $\mathcal{D}$ is compact. For $\Delta \in \mathcal{D}$, put $X = \widehat{J}_\Delta$. The operator X is nonzero, Hermitian, and traceless, so $\lambda_{\max}(-X) > 0$. Along the ray $\Omega_{A,B} + t\Delta$, complete positivity is equivalent to

$$\frac{I}{N} + tX \ge 0.$$

The first boundary point occurs at

$$t_\Delta = \frac{1}{N\lambda_{\max}(-X)}.$$

Every nonzero affine perturbation has a unique representation as a positive multiple of an element of $\mathcal{D}$, and every ray from the relative interior point $\Omega_{A,B}$ meets the relative boundary before leaving the channel body. Consequently,

$$\rho_\diamond(\Omega_{A,B}) = \min_{\Delta \in \mathcal{D}} t_\Delta = \frac{1}{N \max_{\Delta \in \mathcal{D}} \lambda_{\max}(-\widehat{J}_\Delta)}.$$

Continuity on the compact set $\mathcal{D}$ gives attainment and the displayed formula for $\Gamma_{A,B}$.

Lemma 3.1 and (2) give

$$\lambda_{\max}(-X) \le \frac{1}{2}\|X\|_1 \le \frac{1}{2}\|\Delta\|_\diamond,$$

so $\Gamma_{A,B} \le 1/2$. Positivity follows from any nonzero admissible direction, after reversing its sign if necessary. ☐

Remark 4.4 (Effective Schmidt rank). The parameter $s = \min\{d_A, d_B\}$ is the largest Schmidt rank that can be supported simultaneously by input and output. The extremal perturbation is detected by a maximally entangled probe of rank s supported on an s-dimensional input subspace $S_A \subseteq A$; the witness $V_0, V_1 : S_A \to B$ with $\mathrm{Tr} V_0^\dagger V_1 = 0$ of the appendix proof uses only this subspace. Extra input dimensions beyond output capacity do not reduce the radius.

4.3 Benchmark distance

For square systems, the identity channel gives a convenient normalisation check for the Choi bounds and the extremal distance scale used later in the unitary and programming examples.

Corollary 4.5 (Identity versus depolarizing)**.** *For a d-dimensional system,*

$$\|\mathcal{I} - \Omega_d\|_\diamond = 2\left(1 - \frac{1}{d^2}\right).$$

Proof. Let $P = |\omega_H\rangle\langle\omega_H|$. Evaluation on the maximally entangled state gives

$$\|\mathcal{I} - \Omega_d\|_\diamond \ge \left\|P - \frac{I_{H\otimes H}}{d^2}\right\|_1 = 2\left(1 - \frac{1}{d^2}\right).$$

The unnormalized Choi operator is $J = dP - I_{H\otimes H}/d$, and direct spectral decomposition gives

$$\mathrm{Tr}_{H_{\mathrm{out}}} |J| = 2\left(1 - \frac{1}{d^2}\right) I_H,$$

the partial trace over the output copy (the input-copy marginal is the same by symmetry). Lemma 3.2 supplies the matching upper bound. ☐

5 Antipodal sections and the channel lower envelope

The full-body in-radius controls every subcritical affine dimension, but smaller source families can have much larger antipodal separation. This section uses input-independent replacement channels and an input-dependent observable construction to obtain the maximal metric bound, then treats a coherent unitary promise, and finally combines these sections with the full-direction sphere and the full channel ball.

5.1 Replacement-family geometry

The replacement family is an isometric copy of the output state space.

Lemma 5.1 (Centered state radius and replacement isometry)**.** *The centred relative trace-norm in-radius of $\mathcal{S}(B)$ about I_B/d_B is exactly $2/d_B$. Moreover,*

$$\|\mathcal{R}_\sigma - \mathcal{R}_\tau\|_\diamond = \|\sigma - \tau\|_1.$$

Proof. If $X = X^\dagger$, $\operatorname{Tr} X = 0$, and $\|X\|_1 \leq 2/d_B$, then Lemma 3.1 gives $I_B/d_B + X \geq 0$. Conversely, put

$$\tau_\partial = \frac{I_B - |0\rangle\langle 0|}{d_B - 1}, \qquad X = \tau_\partial - \frac{I_B}{d_B}.$$

The direction X has eigenvalue $-1/d_B$ on $|0\rangle$ and eigenvalue $1/[d_B(d_B-1)]$ on its orthogonal complement, so $\|X\|_1 = 2/d_B$. Moreover, for every $\lambda > 1$, the operator $I_B/d_B + \lambda X$ has eigenvalue $(1-\lambda)/d_B < 0$ on $|0\rangle$. Given any radius $R > 2/d_B$, choose $1 < \lambda < Rd_B/2$; then $\|\lambda X\|_1 < R$, while the perturbed operator is not positive. Hence every larger centred ball contains a trace-one Hermitian operator outside the state space, proving maximality of the stated radius.

For $\rho \in \mathcal{S}(A_0 \otimes A)$,

$$(\mathrm{id} \otimes (\mathcal{R}_\sigma - \mathcal{R}_\tau))(\rho) = \operatorname{Tr}_A(\rho) \otimes (\sigma - \tau).$$

Since $\operatorname{Tr}_A(\rho)$ is a state, its trace norm is one. Multiplicativity of trace norm on tensor products therefore gives $\|\sigma - \tau\|_1$, proving the diamond identity. ☐

Because $d_B > 1$, two orthogonal output states define replacement channels at diamond distance two. Since $\|\Phi\|_\diamond = 1$ for any CPTP Φ, $\|\Phi - \Psi\|_\diamond \leq \|\Phi\|_\diamond + \|\Psi\|_\diamond \leq 2$, so the distance between any two channels is at most two. Hence the diamond diameter of $\mathbf{Chan}(A, B)$ is exactly two for $d_B > 1$.

Proposition 5.2 (Replacement-family width equivalence)**.** *Let*

$$\operatorname{Rep}(A, B) = \{\mathcal{R}_\sigma : \sigma \in \mathcal{S}(B)\}.$$

The replacement map is an isometric homeomorphism from $(\mathcal{S}(B), \|\cdot\|_1)$ onto $(\operatorname{Rep}(A, B), \|\cdot\|_\diamond)$. Consequently,

$$\delta_r^{\mathrm{c}}(\operatorname{Rep}(A, B); \operatorname{Rep}(A, B)) = \delta_r^{\mathrm{c}}(\mathcal{S}(B); \mathcal{S}(B)).$$

In particular, this width is at least 1 for $0 \leq r \leq d_B^2 - 2$ and is zero for $r \geq d_B^2 - 1$.

Proof. Lemma 5.1 proves that the replacement map and its inverse on the image preserve distance. Transporting encoders and decoders through this bijection proves equality of the widths. The lower bound follows from the perfectly separated family in Theorem 5.3 and Theorem 2.4; affine coordinates on the trace-one Hermitian hyperplane give exact encoding at dimension $d_B^2 - 1$. ☐

Theorem 5.3 (Perfectly separated replacement family)**.** *There exists a continuous map*

$$h : S^{d_B^2 - 2} \to \operatorname{Rep}(A, B)$$

such that

$$\|h(u) - h(-u)\|_\diamond = 2 \qquad \textit{for every } u.$$

Consequently,

$$\alpha_r(\mathbf{Chan}(A, B)) = 1, \qquad 0 \leq r \leq d_B^2 - 2.$$

Proof. Let $V = \mathrm{Herm}_0(B)$ and choose a Euclidean unit sphere in this $(d_B^2 - 1)$-dimensional real vector space. For X on that sphere, write $X = X_+ - X_-$ and set

$$t_X = \operatorname{Tr} X_+ = \operatorname{Tr} X_- > 0, \qquad \sigma_X = \frac{X_+}{t_X}.$$

Continuous functional calculus makes $X \mapsto X_+$ continuous, and $t_X > 0$ on the sphere, so $X \mapsto \sigma_X$ is continuous. Since $(-X)_+ = X_-$, the states σ_X and σ_{-X} have orthogonal supports and hence trace norm two. Set $h(X) = \mathcal{R}_{\sigma_X}$ and use Lemma 5.1. The profile value is at least one, while the channel diamond diameter gives the reverse inequality. Restriction to coordinate equators gives every smaller index. $\square$

Remark 5.4 (Replacement-channel origin of the small-code obstruction). Every channel in the preceding sphere discards its input and prepares an output state; in particular, it is entanglement-breaking and transmits no input-dependent information. The unit lower bound through $d_B^2 - 2$ therefore does not arise from coherent channel dynamics or from a need to preserve input entanglement. It is already forced by a high-dimensional continuous family of output preparations whose antipodes are perfectly distinguishable. The later Clifford construction is logically distinct because it shows that a related obstruction persists after restricting the source to reversible unitary dynamics.

Proposition 5.5 (Exact replacement-family profile)**.** *The replacement-family profile is*

$$\alpha_r(\mathrm{Rep}(A, B)) = \begin{cases} 1, & 0 \le r \le d_B^2 - 2, \\ 0, & r \ge d_B^2 - 1. \end{cases}$$

Hence its perfect-separation index is exactly $d_B^2 - 2$.

Proof. Theorem 5.3 proves the first range. For $r \ge d_B^2 - 1$, compose any map $S^r \to \mathrm{Rep}(A, B)$ with the inverse replacement isometry and affine coordinates on the $(d_B^2 - 1)$-dimensional state space. Borsuk–Ulam produces an antipodal collision, so every such map has minimum antipodal distance zero. Taking the supremum gives $\alpha_r = 0$. $\square$

The structured partition constructions are stated and proved in Appendix A.5. They are retained as explicit low-rank subfamilies, although Theorem 5.3 provides the stronger unrestricted replacement-channel bound: the same block structure reappears as the output decompositions of Theorem 5.24, whose balanced instances minimise the pinching thresholds, so the partition optimisation records the geometry behind those thresholds.

Figure 2 summarises the orthogonal-block join construction and the replacement isometry.

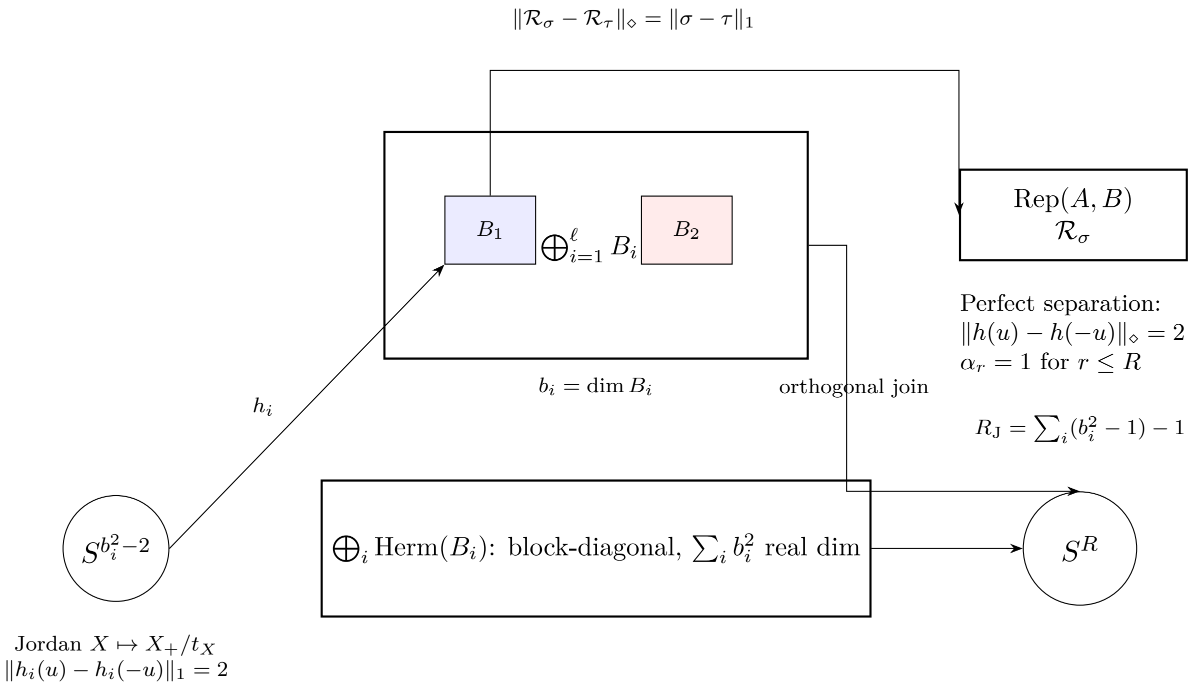
5.2 Unitary Clifford sections

Clifford sections additionally provide a unitary-channel family, whereas the block-join construction of Figure 2 (stated in Appendix A.5) consists of replacement channels. Let

$$\mathbf{UChan}(H) = \{\mathcal{U}_U : X \mapsto UXU^\dagger : U \in \mathrm{U}(H)\}.$$

Proposition 5.6 (Clifford replacement sections)**.** *Let $q = \lfloor \log_2 d_B \rfloor$, $p = 2^q$, and choose a p-dimensional subspace $C \subseteq B$. There is a continuous family of replacement channels on S^{2q} with antipodal diamond distance* 2*.*

Theorem 5.7 (Clifford unitary sections)**.** *If $A = B = H$ is square of dimension $d \ge 2$, let $p = 2^{\lfloor \log_2 d \rfloor}$, choose a p-dimensional subspace $C \subseteq H$, and let H_u denote the anticommuting generators on C of Proposition 5.6. The unitaries $U_u = (I_C + iH_u)/\sqrt{2} \oplus I_{C^\perp}$ give a family in* $\mathbf{UChan}(H)$ *with the same antipodal distance.*



Balanced partitions $Q_\ell(d) = (\ell - t)m^2 + t(m+1)^2$, $d = m\ell + t$, minimise $\sum b_i^2$ and hence the pinching thresholds $r_{\mathrm{out}} = d_A^2(\sum b_j^2 - 1)$ and $r_{\mathrm{in}} = (\sum a_i^2)(d_B^2 - 1)$ of Theorem 5.24. Two-dimensional blocks ($c = 2$) give $\alpha_r = 1$ for $0 \le r \le 3\lfloor d_B/2 \rfloor - 1$ (Corollary A.3).

Figure 2: Replacement-family geometry and orthogonal-block join. Each block B_i carries a sphere $S^{b_i^2-2}$ with antipodal trace norm 2 via the Jordan construction $X \mapsto X_+/\mathrm{Tr}X_+$ of Theorem 5.3, restricted to $\mathrm{Herm}_0(B_i)$. Orthogonal supports make the trace norm additive, giving $\|H(y) - H(-y)\|_1 = \sum_i \lambda_i \|h_i(u_i) - h_i(-u_i)\|_1$, which equals 2 when every block sphere is perfectly separated (Proposition A.1). The replacement isometry (Lemma 5.1) transfers this to diamond distance 2 for channels; the n-outcome analogue for quantum instruments is developed in Ref. [11]. Balanced $Q_\ell(d)$ minimise the pinching thresholds (Corollary 5.26).

The proofs are given in Appendix A.6.

Corollary 5.8 (Unitary-promise obstruction)**.** *If $H \simeq \mathbb{C}^d$ and $q = \lfloor \log_2 d \rfloor$, then*

$$\alpha_r(\mathbf{UChan}(H)) = 1, \qquad 0 \le r \le 2q.$$

Consequently, every continuous encoder of the unitary promise class through an m-dimensional quantum-state alphabet with $m^2 - 1 \le 2q$ has worst-case diamond reconstruction error at least 1, even with an unrestricted decoder.

Proof. Restrict the unitary Clifford sphere in Theorem 5.7 to a coordinate equator S^r. Its antipodes have diamond distance 2, giving the profile lower bound 1; the universal channel-diameter bound gives the reverse inequality. The state-space code embeds into $\mathbb{R}^{m^2-1}$, so Proposition 2.7 and Theorem 2.6 give the memory statement. □

Proposition 5.9 (Unitary-promise zero-error coordinate bound)**.** *Let H have dimension $d > 1$. If a continuous encoder $f : \mathbf{UChan}(H) \to \mathbb{R}^r$ admits an exact decoder on its image, then*

$$r \ge d^2.$$

This is a lower bound on the Euclidean embedding dimension of the unitary promise class; whether d^2 coordinates admit an exact continuous encoder is not addressed here. At $d = 2$

the bound is not sharp: $\mathbf{UChan}(\mathbb{C}^2) \simeq \mathrm{PU}(2) \cong \mathbb{RP}^3$ *embeds in* $\mathbb{R}^5$ *but not in* $\mathbb{R}^4$ *(its minimal embedding dimension is* 5*), so in fact* $r \geq 5$ *there; the sharp embedding dimension of* $\mathrm{PU}(d)$ *for general* d *is not pursued here.*

Proof. Exact reconstruction forces f to be injective. The unitary-channel space is the compact boundaryless manifold $\mathbf{UChan}(H) \simeq \mathrm{PU}(d)$ of real dimension $d^2 - 1$. The invariance of dimension theorem excludes a topological embedding of this manifold into $\mathbb{R}^r$ for $r < d^2 - 1$. If $r = d^2 - 1$, invariance of domain makes the image of every coordinate neighbourhood open in $\mathbb{R}^{d^2-1}$, so the image of the whole manifold is open. It is also compact and nonempty, which is impossible in Euclidean space. These standard invariance-of-dimension and invariance-of-domain results are reviewed, for example, in [19]. □

Corollary 5.10 (Exact width of the Clifford replacement sphere)**.** *Let* $\mathcal{K}_{\mathrm{Cl}} = \{\mathcal{R}_{\sigma_u} : u \in S^{2q}\}$*, where the Clifford generators* H_u*, the projector* P_C*, and the states* $\sigma_u = (P_C + H_u)/p$ *are as in Appendix A.6. Then*

$$\delta_r^{\mathrm{c}}(\mathcal{K}_{\mathrm{Cl}}; \mathbf{Chan}(A, B)) = 1, \qquad 0 \leq r \leq 2q.$$

Proof. The defining sphere has antipodal distance 2, so its profile is 1 on the stated range and Theorem 2.4 gives the lower bound. For the upper bound, use the constant decoder $\mathcal{R}_{P_C/p}$; the operator P_C/p is a state on B supported on C because $\operatorname{Tr} P_C = p$. Since $\|H_u\|_1 = p$ (eigenvalues ± 1 equal multiplicity),

$$\|\mathcal{R}_{\sigma_u} - \mathcal{R}_{P_C/p}\|_\diamond = 1.$$

□

5.3 Observable and full-direction sections

Theorem 5.11 (Observable antipodal sphere)**.** *Assume* $d_A \geq 2$ *and* $d_B \geq 2$*. Then there exists a continuous map* $h : S^{d_A^2-2} \to \mathbf{Chan}(A, B)$ *whose antipodal images have diamond distance* 2*; consequently*

$$\alpha_r(\mathbf{Chan}(A, B)) = 1 \qquad (0 \leq r \leq d_A^2 - 2).$$

This is the $n = 1$ *case of the observable-sphere theorem for* n*-outcome quantum instruments proved in Ref. [11].*

Proof. Fix two orthogonal pure states σ_+, σ_- on B. For $H \in \mathrm{Herm}_0(A)$, $H \neq 0$, set $K(H) = H/\|H\|_\infty$ and $E_\pm(H) = (I_A \pm K(H))/2$. Then $E_\pm(H) \geq 0$ and $E_+(H) + E_-(H) = I_A$, so

$$\Phi^H(X) = \operatorname{Tr}(E_+(H)X)\,\sigma_+ + \operatorname{Tr}(E_-(H)X)\,\sigma_-$$

is a channel. The Hermitian operator $K(H)$ has norm one, so it admits a unit eigenvector $|\psi\rangle$ with eigenvalue $\varepsilon \in \{+1, -1\}$; for that vector one effect has expectation one and the other expectation zero. Since $E_\pm(-H) = E_\mp(H)$, the outputs of Φ^H and Φ^{-H} on the input $|\psi\rangle\langle\psi|$ are σ_ε and $\sigma_{-\varepsilon}$, which are orthogonal, so their trace norm is 2; the diamond distance is therefore exactly 2 by the diameter bound. Normalisation $H \mapsto H/\|H\|_\infty$ is continuous away from zero, hence on the compact unit sphere of $\mathrm{Herm}_0(A) \cong \mathbb{R}^{d_A^2-1}$; the antipodal images thus form a continuous sphere with diamond distance 2, and coordinate equators supply every smaller index. □

Theorem 5.12 (Full-direction antipodal sphere)**.** *Assume* $d_B > 1$*, and let* $D = D_{A,B}$*. For every* $0 \leq r \leq D - 1$ *there exists a continuous map* $h : S^r \to \mathbf{Chan}(A, B)$ *such that*

$$\|h(u) - h(-u)\|_\diamond \geq \frac{2}{d_A} \qquad \forall u \in S^r.$$

Consequently $\alpha_r(\mathbf{Chan}(A, B)) \geq 1/d_A$ *for all* $0 \leq r < D$*. If* $d_A = 1$ *the separation equals* 2*, so* $\alpha_r = 1$ *for all* $r < D_{1,B}$*, matching the exact replacement profile. This is the* $n = 1$ *case of the corresponding instrument theorem in Ref. [11].*

Proof. This is the $n = 1$ specialisation of the full-direction theorem for instruments proved in Ref. [11], obtained by deleting the flag register; the channel-level construction and its complete verification are as follows. Let $\mathcal{V} = \{Z \in \mathrm{Herm}(A_0 \otimes B) : \mathrm{Tr}_B Z = 0\}$ be the direction space, of real dimension D, equipped with a Euclidean norm and unit sphere $S(\mathcal{V}) \cong S^{D-1}$. For $Z \in S(\mathcal{V})$ write the Jordan decomposition $Z = Z_+ - Z_-$; since $\mathrm{Tr}_B Z = 0$, the partial traces of Z_+ and Z_- agree, and $t := \mathrm{Tr}\, Z_+ = \mathrm{Tr}\, Z_-$ is positive: $t = 0$ would give $Z = -Z_- \leq 0$, and $\mathrm{Tr}\, Z = 0$ then forces $Z = 0$. Set $\Theta_\pm = Z_\pm/t$ and $\mu = \mathrm{Tr}_B\, \Theta_+ = \mathrm{Tr}_B\, \Theta_-$; the states Θ_+ and Θ_- have orthogonal supports and common A_0-marginal μ. If $d_A = 1$ set $J^\pm(Z) = \Theta_\pm$. If $d_A \geq 2$, set $\lambda = 1/d_A$, $\nu = (I_{A_0} - \mu)/(d_A - 1) \geq 0$, and $J^\pm(Z) = \lambda\Theta_\pm + (1-\lambda)\nu \otimes (I_B/d_B)$. In both cases $J^\pm(Z) \geq 0$ and $\mathrm{Tr}_B\, J^\pm(Z) = \lambda\mu + (1-\lambda)\nu = I_{A_0}/d_A$, so $J^\pm(Z)$ are normalised Choi operators of channels; let $h(Z)$ be the channel with Choi operator $J^+(Z)$. The remaining verification is as follows: $h(-Z)$ has Choi operator $J^-(Z)$; the antipodal Choi difference $J^+(Z) - J^-(Z) = \lambda(\Theta_+ - \Theta_-)$ has trace norm $2\lambda = 2/d_A$, which by (2) gives the claimed diamond separation; and $Z \mapsto J^+(Z)$ is continuous because $Z \mapsto Z_+$ is continuous and t is bounded away from zero on the compact sphere. Restricting to coordinate equators covers all smaller dimensions r. ☐

Proposition 5.13 (Observable–replacement join)**.** *Assume $d_A \geq 2$ and $d_B \geq 3$. Then*

$$\alpha_r(\mathbf{Chan}(A, B)) = 1 \qquad (0 \leq r \leq d_A^2 + (d_B - 2)^2 - 3).$$

This is the $n = 1$ case of the instrument-level join of Ref. [11]: at $n = 1$ the join threshold $N_{\mathrm{join}} = (d_B - 2)^2 \geq 1$ is equivalent to $d_B \geq 3$, the boundary value $N_{\mathrm{join}} = 1$ (i.e. $d_B = 3$) being the degenerate case in which the replacement component is the empty sphere and the observable component alone realises the full range.

Proof. Let $W_2 = \mathrm{span}\{|\varphi_+\rangle, |\varphi_-\rangle\}$ be the two-dimensional subspace spanned by the orthogonal output states of the observable sphere in Theorem 5.11; every channel in that sphere has output supported on W_2. Let $W_2^\perp$ be its orthogonal complement, of dimension $d_B - 2 \geq 1$. If $d_B = 3$, then $(d_B - 2)^2 - 2 = -1$: the space $\mathrm{Herm}_0(W_2^\perp)$ is trivial, the replacement sphere $S^{(d_B-2)^2-2}$ is empty, and the claimed range $d_A^2 + (d_B - 2)^2 - 3 = d_A^2 - 2$ is that of the observable sphere alone (Theorem 5.11, applicable since $d_B \geq 2$); assume henceforth $d_B \geq 4$, so that $(d_B - 2)^2 - 2 \geq 2$ and the replacement sphere is nonempty. The Jordan construction of Theorem 5.3, restricted to $\mathrm{Herm}_0(W_2^\perp) \subseteq \mathrm{Herm}_0(B)$, gives a continuous map $h_{\mathrm{rep}} : S^{(d_B-2)^2-2} \to \mathcal{S}(W_2^\perp)$ whose antipodes have orthogonal supports; by Lemma 5.1 the corresponding replacement channels have antipodal diamond distance 2. For $y = (y_1, y_2)$ on the join sphere $S^{(d_A^2-2)+((d_B-2)^2-2)+1}$ with $\lambda = \|y_1\|_2^2$, set $u_1 = y_1/\|y_1\|_2$ and $u_2 = y_2/\|y_2\|_2$ where the corresponding blocks are nonzero, and define

$$H(y)(X) = \lambda\, \Phi^{u_1}(X) + (1-\lambda)\, \mathcal{R}_{\tau_{u_2}}(X),$$

setting the corresponding term to zero when its block vanishes. The endpoint cases $\lambda \in \{0, 1\}$ reduce to a single component, at distance 2, so assume $0 < \lambda < 1$. Both summands are channels, so $H(y)$ is a channel; continuity at vanishing blocks holds as in Proposition A.1, since each vanishing term is bounded in diamond norm by the vanishing weight. On the input $|\psi\rangle\langle\psi|$ exposing the pair $(\Phi^{u_1}, \Phi^{-u_1})$, the output of $H(y) - H(-y)$ splits into the part $\lambda(\sigma_\varepsilon - \sigma_{-\varepsilon})$ supported on W_2 and the part $(1-\lambda)(\tau_{u_2} - \tau_{-u_2})$ supported on $W_2^\perp$; the two supports are mutually orthogonal and each part has trace norm 2λ, respectively $2(1-\lambda)$, so the total trace norm is 2, and the diamond distance is exactly 2 by the diameter bound. Restriction to coordinate equators supplies every smaller index, and $\alpha_r \leq 1$ always by the diameter bound. ☐

Theorem 5.14 (Global in-radius and the optimal centred ball)**.** *For every $\Phi_0 \in \mathbf{Chan}(A, B)$,*

$$\rho_\diamond(\Phi_0) \leq \rho_\diamond(\Omega_{A,B}) = \rho_{\mathrm{dep}}(A, B).$$

Thus $\rho_{\mathrm{dep}}(A, B)$ is the largest relative ball contained in the channel body about any centre, and the centred-ball term of the certified envelope cannot be improved by re-centring. The concavity argument is the $n = 1$ case of the instrument-level statement of Ref. [11].

Proof. For a convex body K in a finite-dimensional normed affine space, the relative in-radius about any centre $x \in K$ equals the infimum over closed halfspaces $H \supseteq K$ of the distance from x to ∂H: the ball $B(x, \rho) \subseteq K$ holds if and only if $B(x, \rho) \subseteq H$ for every such H, by separation, and the two infima coincide (at a relative-boundary centre a supporting halfspace contains the centre on its boundary, and both sides vanish). Each map $x \mapsto \operatorname{dist}(x, \partial H)$ is affine, so the in-radius is a pointwise infimum of affine functions, hence concave on K. The in-radius is moreover 1-Lipschitz, hence continuous: $B(x, \rho) \subseteq K$ implies $B(x', \rho - \|x' - x\|) \subseteq K$ by the triangle inequality, so $\rho_\diamond(\Phi_0') \geq \rho_\diamond(\Phi_0) - \|\Phi_0' - \Phi_0\|_\diamond$, and symmetrically.

For $V \in \mathrm{U}(B)$ let $T_V(\Psi) = \mathcal{C}_V \circ \Psi$ with $\mathcal{C}_V(Y) = VYV^\dagger$. This is an affine isometry of $\mathbf{Chan}(A, B)$ by unitary invariance of the diamond norm, so $\rho_\diamond(T_V(\Phi_0)) = \rho_\diamond(\Phi_0)$. The Haar barycentre of Φ_0 under this action is $\Omega_{A,B}$: averaging first over $\mathrm{U}(B)$, Schur's lemma gives $\int V\Psi(X)V^\dagger \, dV = (\operatorname{Tr} \Psi(X)/d_B) I_B$, and $\operatorname{Tr} \Psi(X) = \operatorname{Tr} X$ for the trace-preserving Ψ, so the barycentre maps $X \mapsto \operatorname{Tr}(X) I_B/d_B = \Omega_{A,B}(X)$. Jensen's inequality for the concave continuous in-radius (applied to finite convex combinations of orbit points and extended to the Haar integral by continuity) gives

$$\rho_\diamond(\Omega_{A,B}) = \rho_\diamond\Big(\int T_V(\Phi_0) \, dV \Big) \geq \int \rho_\diamond(T_V(\Phi_0)) \, dV = \rho_\diamond(\Phi_0),$$

as claimed. $\square$

5.4 The channel lower envelope

The replacement, observable, and joined spheres supply the strongest bound at small indices, while the globally optimal centred ball and the full-direction section remain available throughout the full affine range. The following envelope records their pointwise maxima and keeps its positive entries separate from the exact zero-error threshold.

Definition 5.1 (Certified channel lower envelope)**.** Set

$$L_{\mathrm{rep}} = d_B^2 - 2,$$

$$L_{\mathrm{obs}} = \begin{cases} d_A^2 - 2, & d_A \geq 2,\ d_B \geq 2, \\ -1, & \text{otherwise}, \end{cases} \qquad L_{\mathrm{join}} = \begin{cases} d_A^2 + (d_B - 2)^2 - 3, & d_A \geq 2,\ d_B \geq 3, \\ -1, & \text{otherwise}, \end{cases}$$

and

$$L_{A,B} = \min\Big\{ D_{A,B} - 1,\ \max\{L_{\mathrm{rep}}, L_{\mathrm{obs}}, L_{\mathrm{join}}\} \Big\}, \qquad \gamma_{A,B} = \max\Big\{ \rho_{\mathrm{dep}}(A, B),\ \frac{1}{d_A} \Big\}.$$

Then

$$\beta_r(A, B) = \begin{cases} 2 - \rho_{\mathrm{dep}}(A, B), & r = 0, \\ 1, & 1 \leq r \leq L_{A,B}, \\ \gamma_{A,B}, & L_{A,B} < r < D_{A,B}, \\ 0, & r \geq D_{A,B}. \end{cases}$$

When $d_A = 1$ the observable and join terms are vacuous and $L_{1,B} = d_B^2 - 2 = D_{1,B} - 1$, so the middle range is empty. The value at $r = 0$ is the exact constant-code width (Theorems 5.19 and 5.20): the envelope is upgraded there from the one-sphere value 1 to $2 - \rho_{\mathrm{dep}}(A, B)$, so the lower and upper envelopes coincide at $r = 0$. Since $d_B \min\{d_A, d_B\} \geq 2$ gives $2 - \rho_{\mathrm{dep}}(A, B) \geq 1$ and $d_A \geq 1$ gives $0 < \gamma_{A,B} \leq 1$, the displayed envelope is non-increasing. Each component is certified by Theorems 5.3, 5.11, and 5.12, Proposition 5.13, and Theorem 2.8; the ball term $\rho_{\mathrm{dep}}(A, B)$ is the exact centred radius of Theorem 4.1, whose optimality among all centres is Theorem 5.14; the value at $r = 0$ is the exact constant-code width of Theorems 5.19 and 5.20. The quantity $L_{A,B}$ is the $n = 1$ value of the separation index and $\gamma_{A,B}$ the $n = 1$ envelope value of the instrument-level theory of Ref. [11].

Remark 5.15 (Certified envelope). The envelope β_r is a certified lower bound, exact at $r = 0$. The drop from 1 to $\gamma_{A,B}$ at $r = L_{A,B}$ records the certified constructions used here (replacement, observable, and joined spheres; the globally optimal centred ball of Theorem 5.14; the full-direction section), not an exact evaluation of $\alpha_r(\mathbf{Chan}(A, B))$, which may remain larger than $\gamma_{A,B}$ for some $r > L_{A,B}$.

Theorem 5.16 (Channel reconstruction lower envelope)**.** *Let $\mathbf{TP}_{\mathrm{HP}}(A, B)$ be the affine space of Hermiticity-preserving, trace-preserving maps with diamond distance, and define*

$$\widetilde{\delta}_r^{\diamond}(A, B) = \delta_r^{\mathrm{c}}(\mathbf{Chan}(A, B); \mathbf{TP}_{\mathrm{HP}}(A, B)),$$
$$\delta_r^{\diamond}(A, B) = \delta_r^{\mathrm{c}}(\mathbf{Chan}(A, B); \mathbf{Chan}(A, B)).$$

Then

$$\delta_r^{\diamond}(A, B) \geq \widetilde{\delta}_r^{\diamond}(A, B) \geq \alpha_r(\mathbf{Chan}(A, B)) \geq \beta_r(A, B).$$

Moreover,

$$\widetilde{\delta}_r^{\diamond}(A, B) = \delta_r^{\diamond}(A, B) = 0 \qquad (r \geq D_{A,B}).$$

The positive values in β_r are lower bounds; the zero-error threshold is exact. Note that $\widetilde{\delta}_r^{\diamond} \leq \delta_r^{\diamond}$, so vanishing of $\delta_r^{\diamond}$ implies vanishing of $\widetilde{\delta}_r^{\diamond}$.

Proof. Every channel-valued decoder is $\mathbf{TP}_{\mathrm{HP}}(A, B)$-valued, which gives the first inequality. Theorem 2.4 gives the second. At $r = 0$ every encoder is constant, and Theorem 5.19 gives $\delta_0^{\diamond}(A, B) = \widetilde{\delta}_0^{\diamond}(A, B) = 2 - \rho_{\mathrm{dep}}(A, B) = \beta_0(A, B)$, so the bracket is exact at $r = 0$. Theorem 5.3, Theorem 5.11, and Proposition 5.13 give the value one through index $L_{A,B}$. For $L_{A,B} < r < D_{A,B}$, the radial map of Theorem 2.8, applied to the exact centred ball of Theorem 4.1 (optimal among all centres by Theorem 5.14, so re-centring cannot certify more), gives the lower bound $\rho_{\mathrm{dep}}(A, B)$, and Theorem 5.12 gives the lower bound $1/d_A$; the pointwise maximum is $\gamma_{A,B}$. Proposition 2.11 gives exact reconstruction at and above $D_{A,B}$. □

Remark 5.17 (Output preparation and dynamical response). The breakpoint $L_{A,B} + 1$ separates two scales, an input-independent one and an input-dependent one. A single output density operator carries $d_B^2 - 1$ affine parameters, while an input-dependent channel carries $d_A^2(d_B^2 - 1)$. Up to $L_{A,B}$, error one is forced by two effects: preparing and recording an output state alone obstructs the range up to $d_B^2 - 2$, and input-dependent effect readout — the observable sphere (Theorem 5.11) and its join with the replacement sphere (Proposition 5.13) — extends the obstruction whenever $d_A^2 \geq 4d_B - 2$, in particular for every square system of dimension at least four. Beyond $L_{A,B}$, what remains to be resolved is the dependence of the output on the input operator; there the globally optimal centred ball and the full-direction section certify $\gamma_{A,B} = \max\{\rho_{\mathrm{dep}}(A, B), 1/d_A\}$, and for $d_A \leq d_B$ the input-dependent term dominates, so $\gamma_{A,B} = 1/d_A$. The two scales distinguish output-state complexity from input-dependent process response; they do not imply a decomposition theorem for optimal physical encoders.

Definition 5.2 (Perfect-separation indices)**.** Define

$$\pi(A, B) = \sup\{r \in \mathbb{N}_0 : \alpha_r(\mathbf{Chan}(A, B)) = 1\},$$

and

$$\pi_{\mathrm{U}}(d) = \sup\{r \in \mathbb{N}_0 : \alpha_r(\mathbf{UChan}(\mathbb{C}^d)) = 1\}.$$

Corollary 5.18 (Known perfect-separation ranges)**.** *The proved bounds are*

$$L_{A,B} \leq \pi(A, B) \leq D_{A,B} - 1,$$

and, for every $d \geq 2$,

$$2\lfloor \log_2 d \rfloor \leq \pi_{\mathrm{U}}(d) \leq 2d^2 - 3.$$

The replacement-family index is exactly $d_B^2 - 2$ by Proposition 5.5. When $d_A = 1$, the channel body is the output state space, and therefore

$$\pi(A, B) = d_B^2 - 2.$$

Proof. The lower bound on $\pi(A, B)$ follows from Theorems 5.3 and 5.11 and from Proposition 5.13; the unitary bounds follow from Corollary 5.8. For $r \geq D_{A,B}$, affine coordinates and Borsuk–Ulam force an antipodal collision for every map from S^r into the channel body. Hence $\alpha_r = 0$ throughout that range.

The manifold $\mathbf{UChan}(\mathbb{C}^d) \simeq \mathrm{PU}(d)$ has real dimension d^2-1. Whitney's embedding theorem gives an embedding into $\mathbb{R}^{2d^2-2}$ [20]. Proposition 2.7 then gives $\alpha_r(\mathbf{UChan}(\mathbb{C}^d)) = 0$ for $r \geq 2d^2 - 2$, proving the unitary upper bound. □

5.5 Constant reconstruction and Chebyshev geometry

The lower envelope does not determine the positive full-channel widths. A complementary upper bound is obtained from constant reconstruction. For $r = 0$, this is the entire coding problem, so its minimax radius can be evaluated exactly before being used at every larger latent dimension.

Theorem 5.19 (Depolarizing channel as a Chebyshev centre)**.** *For $\Psi \in \mathbf{TP}_{\mathrm{HP}}(A, B)$ define*

$$R(\Psi) = \sup_{\Phi \in \mathbf{Chan}(A,B)} \|\Phi - \Psi\|_\diamond.$$

Then

$$R(\Omega_{A,B}) \leq R(\Psi) \qquad \text{for every } \Psi \in \mathbf{TP}_{\mathrm{HP}}(A, B).$$

Consequently,

$$\widetilde{\delta}_0^\diamond(A, B) = \delta_0^\diamond(A, B) = \sup_{\Phi \in \mathbf{Chan}(A,B)} \|\Phi - \Omega_{A,B}\|_\diamond.$$

Uniqueness of the optimal centre is left open.

Proof. For $V \in \mathrm{U}(B)$, let $\mathcal{C}_V(Y) = VYV^\dagger$ and $T_V(\Psi) = \mathcal{C}_V \circ \Psi$. This is an affine bijection of both $\mathbf{Chan}(A, B)$ and $\mathbf{TP}_{\mathrm{HP}}(A, B)$, and unitary invariance of the diamond norm gives $R(T_V(\Psi)) = R(\Psi)$. The function R is convex because it is a supremum of norm-distance functions.

Let dV be normalised Haar measure and put $\mathcal{T}(Y) = \int VYV^\dagger \, dV$. Haar invariance implies $W\mathcal{T}(Y)W^\dagger = \mathcal{T}(Y)$ for every unitary W on B. Thus $\mathcal{T}(Y)$ is scalar, and trace preservation of the integral fixes the scalar:

$$\mathcal{T}(Y) = \frac{\operatorname{Tr} Y}{d_B} I_B.$$

Therefore every trace-preserving Ψ satisfies

$$\int_{\mathrm{U}(B)} T_V(\Psi)(X) \, dV = \frac{\operatorname{Tr} \Psi(X)}{d_B} I_B = \Omega_{A,B}(X).$$

Jensen's inequality now gives

$$R(\Omega_{A,B}) \leq \int R(T_V(\Psi)) \, dV = R(\Psi).$$

Taking the infimum over $\mathbf{TP}_{\mathrm{HP}}(A, B)$ and over its subset $\mathbf{Chan}(A, B)$ proves the two equalities. □

Theorem 5.20 (Exact Chebyshev covering radius)**.** *Let $s = \min\{d_A, d_B\}$. Then*

$$R(\Omega_{A,B}) = 2\left(1 - \frac{1}{d_B s}\right) = 2 - \rho_{\mathrm{dep}}(A, B).$$

Consequently,

$$\widetilde{\delta}_0^\diamond(A, B) = \delta_0^\diamond(A, B) = 2 - \rho_{\mathrm{dep}}(A, B).$$

The proof is given in Appendix A.2.

Remark 5.21 (Local robustness and maximal departure from noise). The exact formulas give the special identity

$$\rho_\diamond(\Omega_{A,B}) + R(\Omega_{A,B}) = 2.$$

This is not a generic in-radius/covering-radius relation for convex bodies. Here the same effective support size $d_B \min\{d_A, d_B\}$ controls both the nearest loss of complete positivity under a trace-preserving perturbation and the largest entanglement-assisted diamond distance from complete depolarization. The identity therefore links a local robustness property of the noisy centre to a global discrimination extremum of the channel body.

Proposition 5.22 (Fixed-marginal Chebyshev centre). *Let $K \subseteq \mathbf{Chan}(A, B_1 \otimes B_2)$ be compact and invariant under post-composition by the conjugation channel $\mathrm{id}_{B_1} \otimes \mathcal{C}_V$, where $\mathcal{C}_V(Y) = VYV^\dagger$ and $V \in \mathrm{U}(B_2)$. Suppose*

$$\mathrm{Tr}_{B_2} \circ \Phi = \Lambda \qquad (\Phi \in K)$$

for one fixed channel $\Lambda : A \to B_1$. Among Hermiticity-preserving, trace-preserving centres with this marginal, the channel

$$\Phi_0(X) = \Lambda(X) \otimes \frac{I_{B_2}}{d_{B_2}}$$

minimises the worst-case diamond distance to K.

The proof is given in Appendix A.3.

5.6 Constructive intermediate-dimensional upper bounds

The constant decoder gives an upper bound at every latent dimension but does not use the available coordinates. Two additional constructions provide nontrivial upper bounds: affine projection controls fibers near the full affine threshold, while physical pinching retains selected coherent input or output sectors. Figure 2 illustrates the orthogonal-block join and how balanced partitions $Q_\ell(d)$ minimize the pinching thresholds $r_{\mathrm{out}}, r_{\mathrm{in}}$; Figure 3 shows the resulting upper envelope U_r versus the lower envelope β_r for qubits.

Lemma 5.23 (Exact diamond norm of a pinching error). *Let $H = \bigoplus_{j=1}^{\ell} H_j$ be an orthogonal decomposition into nonzero subspaces, with projections P_j, and define*

$$\mathcal{P}_H(X) = \sum_{j=1}^{\ell} P_j X P_j.$$

Then

$$\| \mathrm{id}_H - \mathcal{P}_H \|_\diamond = 2\left(1 - \frac{1}{\ell}\right).$$

Proof. Let $\omega = e^{2\pi i/\ell}$ and set

$$U_m = \sum_{j=1}^{\ell} \omega^{m(j-1)} P_j, \qquad 0 \le m < \ell.$$

The roots-of-unity identity

$$\frac{1}{\ell} \sum_{m=0}^{\ell-1} \omega^{m(i-j)} = \delta_{ij}$$

shows that averaging the conjugation channels gives

$$\mathcal{P}_H = \frac{1}{\ell}\sum_{m=0}^{\ell-1} \mathcal{U}_{U_m}.$$

Hence

$$\| \operatorname{id}_H - \mathcal{P}_H \|_\diamond \le \frac{1}{\ell}\sum_{m=1}^{\ell-1} \| \operatorname{id}_H - \mathcal{U}_{U_m} \|_\diamond \le 2\left(1 - \frac{1}{\ell}\right).$$

For the reverse inequality, choose unit vectors $|e_j\rangle \in H_j$ and put $|e\rangle = \ell^{-1/2}\sum_j |e_j\rangle$. On their span,

$$(\operatorname{id}_H - \mathcal{P}_H)(|e\rangle\langle e|) = \frac{1}{\ell}(J_\ell - I_\ell),$$

where J_ℓ is the all-ones matrix. Its eigenvalues are $(\ell-1)/\ell$ and $-1/\ell$ with multiplicity $\ell - 1$, so its trace norm is $2(1 - 1/\ell)$. Evaluation on this input gives the matching diamond lower bound. $\square$

Theorem 5.24 (Input- and output-pinching achievability)**.** *The following bounds hold.*

1. *Let $B = \bigoplus_{j=1}^{\ell} B_j$, with $b_j = \dim B_j > 0$, and put*

$$r_{\text{out}} = d_A^2\left(\sum_{j=1}^{\ell} b_j^2 - 1\right).$$

Then, for $r \ge r_{\text{out}}$,

$$\widetilde{\delta}_r^\diamond(A, B) \le \delta_r^\diamond(A, B) \le 2\left(1 - \frac{1}{\ell}\right).$$

2. *Let $A = \bigoplus_{i=1}^{k} A_i$, with $a_i = \dim A_i > 0$, and put*

$$r_{\text{in}} = \left(\sum_{i=1}^{k} a_i^2\right)(d_B^2 - 1).$$

Then, for $r \ge r_{\text{in}}$,

$$\widetilde{\delta}_r^\diamond(A, B) \le \delta_r^\diamond(A, B) \le 2\left(1 - \frac{1}{k}\right).$$

In both cases the decoder may be chosen channel-valued.

Proof. For output pinching, let $\mathcal{P}_B$ be the channel from Lemma 5.23 and set

$$K_{\text{out}} = \{\Psi \in \mathbf{Chan}(A, B) : \mathcal{P}_B \circ \Psi = \Psi\}.$$

The normalised Choi operators of this family are block diagonal in the output decomposition. Their ambient real vector space has dimension $d_A^2 \sum_j b_j^2$. The marginal map

$$\bigoplus_j \operatorname{Herm}(A_0 \otimes B_j) \longrightarrow \operatorname{Herm}(A_0), \qquad (X_j)_j \longmapsto \sum_j \operatorname{Tr}_{B_j} X_j,$$

is surjective: for $H \in \operatorname{Herm}(A_0)$, set $X_1 = H \otimes I_{B_1}/b_1$ and all other blocks to zero. Hence trace preservation imposes d_A^2 independent affine equations. The depolarizing channel belongs to the family and has positive-definite Choi operator. Thus $\dim_{\text{aff}} K_{\text{out}} = r_{\text{out}}$.

Choose affine coordinates on this family and encode $\mathcal{P}_B \circ \Phi$. Invert the coordinates on the encoded image and use any fixed channel, e.g., $\Omega_{A,B}$, elsewhere; since no decoder regularity is required, this is admissible. This gives a continuous encoder and a channel-valued decoder satisfying

$$\begin{aligned}\|\Phi - \mathcal{P}_B \circ \Phi\|_\diamond &\le \|\operatorname{id}_B - \mathcal{P}_B\|_\diamond \|\Phi\|_\diamond \\ &= 2\left(1 - \frac{1}{\ell}\right).\end{aligned}$$

Padding by zero coordinates handles every $r \ge r_{\mathrm{out}}$.

For input pinching, let $\mathcal{P}_A$ be the corresponding channel on A. On the Choi reference, use the basis-conjugate decomposition $A_0 = \bigoplus_i \overline{A}_i$. The Choi operators of channels satisfying $\Psi \circ \mathcal{P}_A = \Psi$ are block diagonal in these input blocks. The ambient block space has dimension $d_B^2 \sum_i a_i^2$. The blockwise marginal map is surjective by placing $H_i \otimes I_B/d_B$ in each prescribed input block, so the constraints have total rank $\sum_i a_i^2$. The depolarizing channel is again a positive-definite relative interior point, giving affine dimension r_{in}.

Encode $\Phi \circ \mathcal{P}_A$ by affine coordinates and invert on the image, defining it as $\Omega_{A,B}$ outside the image. Submultiplicativity and Lemma 5.23 give

$$\|\Phi - \Phi \circ \mathcal{P}_A\|_\diamond \le \|\operatorname{id}_A - \mathcal{P}_A\|_\diamond = 2\left(1 - \frac{1}{k}\right).$$

This proves the second claim. □

Lemma 5.25 (Balanced partitions minimise squared block sums)**.** *For $1 \le \ell \le d$, write $d = m\ell + t$, $0 \le t < \ell$. Among all partitions of d into ℓ positive parts, the sum of squared parts is minimised by the balanced partition with $\ell - t$ parts equal to m and t parts equal to $m+1$, and the minimum is*

$$Q_\ell(d) = (\ell - t)m^2 + t(m+1)^2.$$

Proof. If two block sizes satisfy $a \ge b + 2$, moving one dimension from the larger to the smaller changes their squared sum by $-2(a - b - 1) < 0$; each exchange strictly decreases the sum and there are only finitely many partitions of d into ℓ positive parts, so iteration terminates at the balanced partition. □

Corollary 5.26 (Balanced pinching bounds)**.** *For $1 \le \ell \le d_B$,*

$$\delta_r^\diamond(A,B) \le 2\left(1 - \frac{1}{\ell}\right) \quad \text{if} \quad r \ge d_A^2(Q_\ell(d_B) - 1).$$

For $1 \le k \le d_A$,

$$\delta_r^\diamond(A,B) \le 2\left(1 - \frac{1}{k}\right) \quad \text{if} \quad r \ge Q_k(d_A)(d_B^2 - 1).$$

The same bounds hold for $\widetilde{\delta}_r^\diamond(A,B)$.

Proof. Choose balanced block decompositions in Theorem 5.24. Lemma 5.25 shows that they minimise the required affine dimension for a fixed number of blocks. □

In particular, a balanced two-sector output pinching gives

$$\delta_r^\diamond(A,B) \le 1 \quad \text{for} \quad r \ge d_A^2\left(\left\lfloor \frac{d_B}{2} \right\rfloor^2 + \left\lceil \frac{d_B}{2} \right\rceil^2 - 1\right),$$

while complete output pinching gives

$$\delta_r^\diamond(A,B) \le 2\left(1 - \frac{1}{d_B}\right) \quad \text{for} \quad r \ge d_A^2(d_B - 1).$$

Remark 5.27 (Operational meaning of pinching). Output pinching is fixed CPTP postprocessing and preserves every observable that is block diagonal in the selected output sectors. Input pinching is fixed CPTP preprocessing and preserves the channel action on every block-diagonal input state. The diamond bounds quantify the worst loss of inter-sector coherence under arbitrary entangled probes and output measurements. The subsequent affine-coordinate decoder remains the unrestricted representational decoder, not a fixed programmable processor.

Definition 5.3 (Constructive channel upper envelope)**.** For $0 \le r \le D_{A,B}$, let $U_r(A,B)$ be the minimum of

$$2 - \rho_{\mathrm{dep}}(A,B), \qquad \frac{2(D_{A,B} - r)}{D_{A,B} - r + 1},$$

all values $2(1 - 1/\ell)$ with $1 \le \ell \le d_B$ for which $r \ge d_A^2(Q_\ell(d_B) - 1)$, and all values $2(1 - 1/k)$ with $1 \le k \le d_A$ for which $r \ge Q_k(d_A)(d_B^2 - 1)$. A minimum over an empty pinching family is ignored (in particular $\ell = 1$ or $k = 1$ gives value 0 at $r \ge D_{A,B}$).

Corollary 5.28 (Error-one floor of the constructive upper envelope)**.** *For $0 \le r < D_{A,B}$, every admissible term of Definition 5.3 is at least 1, hence $U_r(A,B) \ge 1$.*

Proof. The constant-code term satisfies $2 - \rho_{\mathrm{dep}}(A,B) \ge 1$ because $d_B \min\{d_A, d_B\} \ge 2$. The convex-projection term satisfies $2(D_{A,B} - r)/(D_{A,B} - r + 1) \ge 1$ because $D_{A,B} - r \ge 1$, with equality only at $r = D_{A,B} - 1$. Output pinching gives $2(1 - 1/\ell) \ge 1$ whenever $\ell \ge 2$, while $\ell = 1$ requires $r \ge d_A^2(Q_1(d_B) - 1) = D_{A,B}$; similarly $k \ge 2$ below $D_{A,B}$, because $Q_1(d_A)(d_B^2 - 1) = D_{A,B}$. ☐

Corollary 5.29 (Exact top-subcritical replacement width)**.** *The replacement-channel family satisfies*

$$\delta^{\mathrm{c}}_{d_B^2 - 2}(\mathrm{Rep}(A,B); \mathrm{Rep}(A,B)) = 1.$$

Proof. The lower bound follows from Proposition 5.5 and Theorem 2.4. The replacement family is a compact convex body of affine dimension $d_B^2 - 1$ and diameter two. Proposition 2.10, applied at $r = d_B^2 - 2$, gives the matching upper bound. ☐

Corollary 5.30 (Rigorous interval for full-channel widths)**.** *For $0 \le r < D_{A,B}$,*

$$\beta_r(A,B) \le \alpha_r(\mathbf{Chan}(A,B)) \le \widetilde{\delta}^{\diamond}_r(A,B) \le \delta^{\diamond}_r(A,B) \le U_r(A,B).$$

At $r = 0$, the two widths and both envelopes equal $U_0(A,B) = 2 - \rho_{\mathrm{dep}}(A,B)$. For $r \ge D_{A,B}$, both widths are zero.

Proof. The lower inequalities are Theorem 5.16. The four classes of upper bounds in Definition 5.3 follow from Theorem 5.20, Proposition 2.10, and Corollary 5.26. Their minimum is therefore an admissible upper bound. At $r = 0$, the convex-projection term is $2D_{A,B}/(D_{A,B} + 1)$. Since

$$D_{A,B} + 1 = d_A^2(d_B^2 - 1) + 1 \ge d_B \min\{d_A, d_B\},$$

indeed, if $d_A = 1$ then $D_{A,B} + 1 = d_B^2 \ge d_B = d_B \min\{d_A, d_B\}$; if $d_A \ge 2$ then $D_{A,B} + 1 \ge 2(d_B^2 - 1) + 1 = 2d_B^2 - 1 \ge d_B^2 \ge d_B \min\{d_A, d_B\}$. Hence this term is at least $2 - \rho_{\mathrm{dep}}(A,B)$, and every nontrivial pinching threshold is positive. The exact constant-code theorem therefore gives $U_0(A,B) = 2 - \rho_{\mathrm{dep}}(A,B)$ and equality of both widths and both envelopes at $r = 0$. Proposition 2.11 gives zero width at and above $D_{A,B}$. ☐

Remark 5.31 (Exact and non-exact quantities). The positive reconstruction widths of the full channel body are not determined here. Exact results are obtained for the fiber-radius characterization, the replacement-family profile, the Clifford replacement sphere, the replacement and full-channel zero-error thresholds, the all-dimensional depolarizing-centred in-radius, the exact

Source geometry or regime	Index range	Guarantee	Status
Replacement sphere	$0 \le r \le d_B^2 - 2$	$\alpha_r = 1$	Exact profile value
Observable sphere	$0 \le r \le d_A^2 - 2$ $(d_A, d_B \ge 2)$	$\alpha_r = 1$	Exact profile value
Observable–replacement join	$0 \le r \le d_A^2 + (d_B - 2)^2 - 3$ $(d_A \ge 2,\, d_B \ge 3)$	$\alpha_r = 1$	Exact profile value
Full channel ball	$L_{A,B} < r < D_{A,B}$	$\alpha_r \ge \gamma_{A,B}$	Lower bound for both widths, $\gamma_{A,B} = \max\{\rho_{\mathrm{dep}}(A,B),\ 1/d_A\}$
Global in-radius	centred geometry	$\rho_\diamond(\Omega_{A,B}) = \rho_{\mathrm{dep}}(A,B)$ is optimal over all centres	Exact centred radius
Euclidean channel code	$r \ge D_{A,B}$	$\widetilde{\delta}_r^\diamond = \delta_r^\diamond = 0$	Exact abstract threshold
Constant code	$r = 0$	$\widetilde{\delta}_0^\diamond = \delta_0^\diamond = 2 - \rho_{\mathrm{dep}}(A,B)$	Exact covering radius
Convex projection	$0 \le r \le D_{A,B}$	$\delta_r^\diamond \le 2(D_{A,B} - r)/(D_{A,B} - r + 1)$	Upper bound
Balanced block pinching	partition-dependent thresholds	$\delta_r^\diamond \le 2(1 - 1/\ell)$	Constructive upper bound

Table 1: Exact statements and constructive bounds. The positive full-body reconstruction widths below $D_{A,B}$ remain open and are bracketed by the two envelopes.

constant-code covering radius, and the top-subcritical replacement-family width. The convex-projection and block-pinching results give explicit upper bounds, not exact positive full-body widths.

Table 1 summarises the exact quantities and the certified lower and constructive upper bounds.

Figure 3 visualizes Table 1 for the qubit case.

Example 5.32 (Qubit channels). For $d_A = d_B = 2$, $D_{2,2} = 12$ and

$$\delta_0^\diamond(2,2) = \frac{3}{2}.$$

For positive subcritical indices,

$$1 \le \widetilde{\delta}_r^\diamond(2,2) \le \delta_r^\diamond(2,2) \le \frac{3}{2} \quad (1 \le r \le 2),$$

while

$$\frac{1}{2} \le \widetilde{\delta}_3^\diamond(2,2) \le \delta_3^\diamond(2,2) \le \frac{3}{2}.$$

The balanced two-sector output-pinching threshold is $r = 4$, so it does not improve the $r = 3$ bound. From that threshold onward, output pinching gives

$$\frac{1}{2} \le \widetilde{\delta}_r^\diamond(2,2) \le \delta_r^\diamond(2,2) \le 1 \quad (4 \le r \le 11).$$

Both widths vanish for $r \ge 12$. For qubits the observable sphere contributes $L_{\mathrm{obs}} = 2 = d_B^2 - 2$, an instance of Corollary 5.30, the join is invalid ($d_B = 2$), and $\gamma_{2,2} = \max\{\rho_{\mathrm{dep}}(2,2), 1/2\} = 1/2$ coincides with $\rho_{\mathrm{dep}}(2,2)$, so the envelope values displayed are unchanged; for $d_A \le d_B$ with $d_B \ge 3$ the input-dependent term $1/d_A$ is strictly larger than $\rho_{\mathrm{dep}}(A,B)$, and for square systems of dimension at least four the join extends the value-one range beyond $d_B^2 - 2$. For qubits the

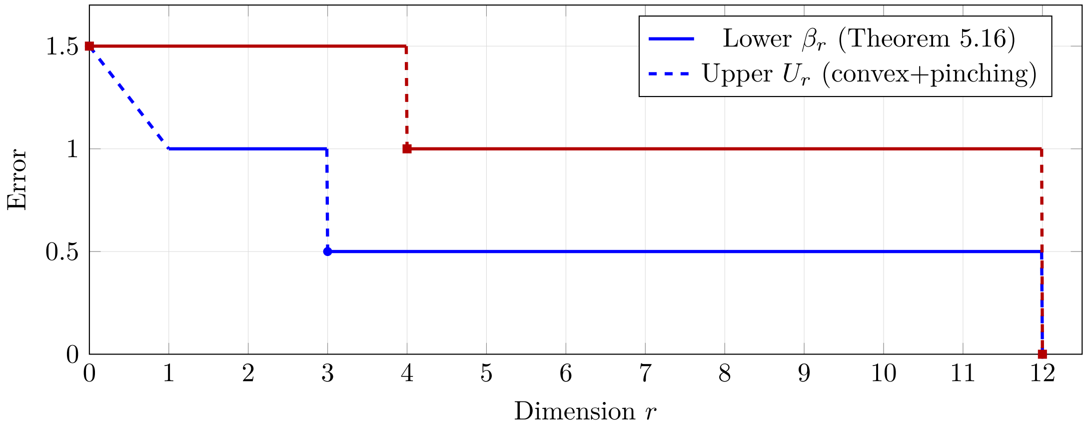


Figure 3: Qubit channels $d_A = d_B = 2$, $D_{2,2} = 12$: certified lower envelope β_r (blue, 3/2 at $r = 0$ — the exact constant-code width, where the envelopes coincide — 1 for $1 \leq r \leq 2$, 0.5 for $3 \leq r \leq 11$, 0 at $r \geq 12$) and constructive upper envelope U_r (red, 1.5 for $0 \leq r \leq 3$, 1 for $4 \leq r \leq 11$, 0 at 12). The drop at $r = 4$ is balanced two-sector output pinching $Q_2(2) = 2$, $r_{\text{out}} = 4$. The gap 0.5 vs 1 for $4 \leq r \leq 11$ is the undetermined positive width interval. For $r = 3$ convex projection gives $2 \cdot 9/10 = 1.8$ but constant decoder 1.5 is tighter.

replacement sphere can be visualized on the Bloch ball: $X \mapsto X_+/t_X$ sends a traceless Hermitian X to the pure state supported on its positive part; X and $-X$ give orthogonal pure states, i.e., antipodal Bloch vectors, with trace norm 2. The replacement isometry transfers this to diamond distance 2. The vanishing of both widths at $r \geq 12$ records the existence of an exact abstract code, not of a physical programming protocol. Figure 3 plots β_r and U_r for this case, showing the coincidence of the two envelopes at $r = 0$ at the exact constant-code width 3/2, the $r = 4$ pinching drop, and the remaining gap.

6 Latent codes and physical programs

The source bounds now pass to concrete code alphabets. Unrestricted Euclidean and quantum-state codes provide the abstract benchmark. Classical–quantum and reachable subsets then refine the relevant embedding dimension, after which the decoder is restricted to a fixed physical processor.

6.1 Abstract latent and quantum-state codes

Theorem 6.1 (General latent-space channel bound)**.** *Let Y be Hausdorff and let the encoder $f : \mathbf{Chan}(A, B) \to Y$ be continuous. If r exceeds* $\operatorname{coind}_{\mathbb{Z}_2} F_2(Y)$*, then every decoder $g : Y \to \mathbf{TP}_{\mathrm{HP}}(A, B)$ satisfies*

$$\sup_{\Phi \in \mathbf{Chan}(A,B)} \|g(f(\Phi)) - \Phi\|_\diamond \geq \alpha_r(\mathbf{Chan}(A, B)).$$

Proof. Take $K = \mathbf{Chan}(A, B)$ and $X = \mathbf{TP}_{\mathrm{HP}}(A, B)$ with the diamond metric in Theorem 2.6. Its coindex hypothesis is exactly the one stated here, and its conclusion is the displayed inequality. □

Theorem 6.2 (Quantum-state latent-code hierarchy)**.** *Let M be an m-dimensional Hilbert space, set $r = m^2 - 1$, and let* $\mathrm{Enc} : \mathbf{Chan}(A, B) \to \mathcal{S}(M)$ *be continuous. For every decoder* $\mathrm{Dec} : \mathcal{S}(M) \to \mathbf{TP}_{\mathrm{HP}}(A, B)$ *(no regularity assumed), one has*

$$\sup_{\Phi \in \mathbf{Chan}(A,B)} \| \mathrm{Dec}(\mathrm{Enc}(\Phi)) - \Phi\|_\diamond \geq \beta_r(A, B).$$

Since channel-valued decoders form a subclass of $\mathbf{TP}_{\mathrm{HP}}(A,B)$-valued decoders, the same lower bound applies a fortiori when the decoder is required to output a channel. If the retained code is known to be classical–quantum, the sharper affine dimension in Proposition 6.5 should be used instead of the dimension of an unrestricted density-operator space.

Proof. The trace-one affine hyperplane of $\mathrm{Herm}(M)$ has dimension $r = m^2 - 1$. If $m = 1$, the state space is a singleton, its deleted product has coindex -1, and Theorem 2.6 applies with $r = 0$. Theorem 5.3 gives $\alpha_0(\mathbf{Chan}(A,B)) = 1 = \beta_0(A,B)$. If $m \geq 2$, affine coordinates inject $\mathcal{S}(M)$ into $\mathbb{R}^r$; Proposition 2.7 therefore bounds its deleted-product coindex by $r-1$, and Theorem 2.6 again gives the profile bound at index r. Theorem 5.16 then yields $\alpha_r(\mathbf{Chan}(A,B)) \geq \beta_r(A,B)$ in both cases. ☐

Proposition 6.3 (Exact abstract memory threshold)**.** *An exact continuous encoding into $\mathcal{S}(M)$ followed by an unrestricted channel-valued decoder exists if and only if*

$$m^2 - 1 \geq D_{A,B}.$$

Thus the least Hilbert-space dimension in this abstract model is

$$m_{\min} = \left\lceil \sqrt{D_{A,B} + 1} \right\rceil .$$

Proof. Necessity follows from Proposition 2.11: the depolarizing channel has positive-definite normalised Choi operator, hence is a relative interior point, so the channel body contains a positive relative ball. For sufficiency, choose affine coordinates $c : \operatorname{aff}\mathbf{Chan}(A,B) \to \mathbb{R}^{D_{A,B}}$ with $c(\Omega_{A,B}) = 0$, and an injective linear map $L : \mathbb{R}^{D_{A,B}} \to \mathrm{Herm}_0(M)$. Put

$$C = \sup_{\Phi} \|L(c(\Phi))\|_\infty < \infty$$

and choose $0 < \varepsilon < 1/(m \max\{1, C\})$. Then

$$\mathrm{Enc}(\Phi) = \frac{I_M}{m} + \varepsilon L(c(\Phi)).$$

The perturbation is traceless and has operator norm strictly below $1/m$ (strict inequality ensures positive definite, not merely semidefinite), so every encoded operator is positive definite with trace one. The map is a continuous affine injection into $\mathcal{S}(M)$. Invert it set-theoretically on its image and define the decoder arbitrarily elsewhere. No continuity or physical implementability of this decoder is asserted. ☐

Corollary 6.4 (Output-sized memory necessity)**.** *For a complete retained quantum-state alphabet $\mathcal{S}(M)$, worst-case error strictly below 1 requires*

$$m \geq d_B.$$

Worst-case error strictly below $\rho_{\mathrm{dep}}(A,B)$ requires

$$m \geq \left\lceil \sqrt{D_{A,B} + 1} \right\rceil .$$

The first condition is only necessary. The second is also sufficient for exact reconstruction in the abstract unrestricted-decoder model by Proposition 6.3; no sufficiency for a fixed physical processor is implied.

Proof. If $m < d_B$, then $m^2 - 1 \leq d_B^2 - 2$, so Theorem 6.2 gives error at least one. If $m^2 - 1 < D_{A,B}$, the same theorem gives error at least $\rho_{\mathrm{dep}}(A,B)$. Taking contrapositives proves both claims. ☐

6.2 Classical–quantum and reachable codes

Proposition 6.5 (Classical–quantum retained code)**.** *Let the retained code consist of block-diagonal states*

$$\bigoplus_{j=1}^{c} \rho_j, \qquad \rho_j \geq 0, \qquad \sum_j \operatorname{Tr} \rho_j = 1,$$

where the jth quantum block acts on a space of dimension m_j. Its affine dimension is

$$\sum_{j=1}^{c} m_j^2 - 1.$$

Consequently, all Euclidean lower bounds apply with this value of r. For equal block dimensions $m_j = m$, it reduces to $cm^2 - 1$.

Proof. A Hermitian block of size m_j has real dimension m_j^2, so the direct sum of all blocks has dimension $\sum_j m_j^2$. The total-trace-one constraint removes one affine degree of freedom. The relative interior consists of those block-diagonal states $\bigoplus_j \rho_j$ for which every block ρ_j is positive definite on its prescribed block space (block traces automatically positive); this set is nonempty, e.g., $\rho_j = I_{m_j}/(cm_j)$. Affine coordinates therefore give the required Euclidean injection. ☐

Corollary 6.6 (Classical–quantum output-size necessity)**.** *For a complete retained classical–quantum alphabet with block dimensions $m_1, \ldots, m_c$, worst-case error strictly below one requires*

$$\sum_{j=1}^{c} m_j^2 \geq d_B^2.$$

For equal block dimensions this becomes $cm^2 \geq d_B^2$.

Proof. Proposition 6.5 embeds the complete alphabet into $\mathbb{R}^{\sum_j m_j^2 - 1}$. If that dimension is at most $d_B^2 - 2$, Theorem 5.3 and the Euclidean source–latent obstruction give error at least one. Taking the contrapositive proves the claim. ☐

Definition 6.1 (Embedding dimension)**.** For a topological space Z, let

$$\operatorname{embdim}(Z) = \inf\{s \in \mathbb{N}_0 : Z \text{ admits a continuous injection into } \mathbb{R}^s\},$$

with value $+\infty$ if none exists.

Proposition 6.7 (Reachable-code refinement)**.** *Let $f : \mathbf{Chan}(A, B) \to Y$ be continuous and put $Z = f(\mathbf{Chan}(A, B))$. If* $\operatorname{embdim}(Z) \leq s$*, then every decoder $g : Y \to \mathbf{TP}_{\mathrm{HP}}(A, B)$ has worst-case diamond error at least $\alpha_s(\mathbf{Chan}(A, B))$. The same lower bound holds for channel-valued decoders.*

Proof. Let $\iota : Z \to \mathbb{R}^s$ be a continuous injection. For every $h : S^s \to \mathbf{Chan}(A, B)$, Borsuk–Ulam applied to $\iota \circ f \circ h$ gives u with $\iota(f(h(u))) = \iota(f(h(-u)))$. Injectivity of ι implies that the original latent codes coincide. The triangle-inequality argument in Theorem 2.6 then gives the lower bound $\alpha_s(\mathbf{Chan}(A, B))$ for every decoder g. ☐

Proposition 6.8 (Structured latent-space coindex bounds)**.** *The following statements hold.*

1. *A singleton has deleted-product coindex -1; a finite discrete space with at least two points has coindex 0; and*

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(S^n) = n.$$

2. *If $n \geq 1$, Y contains a subset homeomorphic to a nonempty open subset of an n-manifold, and Y injects continuously into $\mathbb{R}^s$, then*

$$n - 1 \leq \operatorname{coind}_{\mathbb{Z}_2} F_2(Y) \leq s - 1.$$

3. *Let $\mathcal{S}_k(\mathbb{C}^m)$ denote the manifold of density operators on $\mathbb{C}^m$ of rank exactly k (i.e., $\{\rho \in \mathcal{S}(\mathbb{C}^m) : \operatorname{rank}\rho = k\}$), with its subspace topology. It is Hausdorff but noncompact (and not closed) for $1 \leq k < m$, and the coindex of Definition 2.4 requires only Hausdorffness. For pure states and for $1 \leq k < m$,*

$$2m - 3 \leq \operatorname{coind}_{\mathbb{Z}_2} F_2(\mathbb{CP}^{m-1}) \leq m^2 - 2,$$
$$2mk - k^2 - 2 \leq \operatorname{coind}_{\mathbb{Z}_2} F_2(\mathcal{S}_k(\mathbb{C}^m)) \leq m^2 - 2.$$

For $m = 2$, the pure-state value is exactly 2.

Proof. For a finite discrete space, an equivariant map from S^0 exists, whereas connected spheres cannot map equivariantly into a discrete deleted product. For S^n, the map $u \mapsto (u, -u)$ gives the lower bound and the standard embedding into $\mathbb{R}^{n+1}$ gives the upper bound.

A nonempty open subset of an n-manifold, with $n \geq 1$, contains a coordinate ball B^n and hence a small embedded S^{n-1} as $\{x : \|x\| = \varepsilon\}$, giving the lower bound in the second claim; Proposition 2.7 gives the upper bound. The pure-state manifold has dimension $2m - 2$, and the rank-k stratum has dimension $2mk - k^2 - 1$. Both lie in the $(m^2 - 1)$-dimensional affine space of trace-one Hermitian matrices. Substitution gives the displayed bounds. Finally, $\mathbb{CP}^1 \simeq S^2$. $\square$

Remark 6.9 (Convex code spaces). Proposition 2.7 gives the exact values

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(\mathcal{S}(M)) = m^2 - 2, \qquad \operatorname{coind}_{\mathbb{Z}_2} F_2(\mathbf{Chan}(A, B)) = D_{A,B} - 1,$$

and

$$\operatorname{coind}_{\mathbb{Z}_2} F_2(\operatorname{Rep}(A, B)) = d_B^2 - 2.$$

These equalities use convexity and nonempty relative interior; the nonconvex spaces in Proposition 6.8 generally have only upper and lower bounds.

6.3 Physical programmable processors

The preceding results treat a density operator as an exact latent symbol. A physical programmable decoder is more restrictive. Optimal asymptotic program-cost scaling for approximate universal unitary programming is known [21]. The minimax quantities below isolate fixed finite program dimension and give compactness, separation, and elementary finite-packing consequences in diamond norm.

Let a fixed completely positive trace-preserving (CPTP) map $\mathcal{P} : \mathcal{L}(M \otimes A) \to \mathcal{L}(B)$ induce $\mathcal{P}_\rho(X) = \mathcal{P}(\rho \otimes X)$.

Proposition 6.10 (Physical processor stability and scope)**.** *For all program states,*

$$\|\mathcal{P}_\rho - \mathcal{P}_\sigma\|_\diamond \leq \|\rho - \sigma\|_1.$$

Since the lower bounds are universal over all decoders, they remain valid when the decoder is restricted to a fixed physical processor. This does not mean that the abstract zero-error decoder in Proposition 6.3 can be realized by such a processor: no finite-dimensional deterministic processor exactly programs all unitary channels on a nontrivial system.

Proof. For any state ξ on a reference system tensored with the channel input A, complete trace-norm contractivity gives

$$\|(\mathrm{id} \otimes \mathcal{P})((\rho - \sigma) \otimes \xi)\|_1 \le \|\rho - \sigma\|_1 \|\xi\|_1 = \|\rho - \sigma\|_1.$$

Take the supremum over ξ. The exact universal-programming exclusion is the no-programming theorem [1]. $\square$

Lemma 6.11 (Orthogonality of exact mixed program supports)**.** *Suppose program states ρ and σ exactly implement two distinct unitary channels through one deterministic processor. Then*

$$\operatorname{supp} \rho \perp \operatorname{supp} \sigma.$$

Proof. Write $\rho = \sum_j p_j |a_j\rangle\langle a_j|$ with $p_j > 0$. The induced unitary channel is a convex combination of the channels programmed by the pure states $|a_j\rangle$. A unitary channel is an extreme point of the convex set of channels (its normalised Choi operator is rank one, and the rank-one states are extreme), so every term of positive weight programs that same unitary. The same holds for every eigenvector in the support of σ. The pure-program no-programming theorem [1] makes pure programs for distinct unitary channels orthogonal, proving orthogonality of the two supports. $\square$

Theorem 6.12 (Positive finite-programming lower bound)**.** *Let H have dimension $d > 1$ and let the complete retained program system M have fixed finite dimension m, with no additional side information supplied to the processor. Define*

$$\gamma_{m,d} = \inf_{\mathcal{P}} \sup_{\mathcal{U} \in \mathbf{UChan}(H)} \min_{\rho \in \mathcal{S}(M)} \|\mathcal{P}_\rho - \mathcal{U}\|_\diamond,$$

where the first infimum ranges over CPTP processors $\mathcal{P} : \mathcal{L}(M \otimes H) \to \mathcal{L}(H)$. Then

$$\gamma_{m,d} > 0.$$

Consequently, no number of queries used to prepare such an m-dimensional program can make the worst-case physical programming error tend to zero while m remains fixed, because any finite-query encoder produces some program state in the fixed m-dimensional space and $\gamma_{m,d}$ already takes infimum over all such states and processors.

Proof. Via normalised Choi operators, the processors form a compact subset of a common finite-dimensional matrix space. Indeed, the set of CPTP maps $\mathcal{L}(M \otimes H) \to \mathcal{L}(H)$ is closed (Choi positivity and the linear trace-preserving constraints) and bounded in trace, hence compact. The program-state and unitary-channel sets are compact, and

$$F(\mathcal{P}, \rho, \mathcal{U}) = \|\mathcal{P}_\rho - \mathcal{U}\|_\diamond$$

is jointly continuous. Define

$$\omega(\mathcal{P}, \mathcal{Q}) = \sup_{\rho, \mathcal{U}} |F(\mathcal{P}, \rho, \mathcal{U}) - F(\mathcal{Q}, \rho, \mathcal{U})|.$$

Uniform continuity on the compact program–unitary product gives $\omega(\mathcal{P}, \mathcal{Q}) \to 0$ as $\mathcal{Q} \to \mathcal{P}$. Taking a minimum over ρ and then a maximum over $\mathcal{U}$ changes a function by no more than its uniform perturbation. Therefore

$$e(\mathcal{P}) = \max_{\mathcal{U} \in \mathbf{UChan}(H)} \min_{\rho \in \mathcal{S}(M)} F(\mathcal{P}, \rho, \mathcal{U})$$

is continuous. Hence e attains its minimum over processors. If that minimum were zero, one finite-dimensional processor would exactly implement every unitary channel. Choosing $m + 1$ distinct unitary channels would then produce $m + 1$ nonzero program states with pairwise orthogonal supports by Lemma 6.11, which is impossible in an m-dimensional program space. Any program produced by a finite-query encoder is among the states in the inner optimization, which proves the final claim. $\square$

Proposition 6.13 (Program-state separation). *Suppose a fixed processor approximates channels $\Phi_1, \dots, \Phi_L$ with program states $\rho_1, \dots, \rho_L$ and error at most ε:*

$$\|\mathcal{P}_{\rho_j} - \Phi_j\|_\diamond \le \varepsilon.$$

Then

$$\|\rho_i - \rho_j\|_1 \ge \|\Phi_i - \Phi_j\|_\diamond - 2\varepsilon.$$

Proof. The triangle inequality and Proposition 6.10 give

$$\|\Phi_i - \Phi_j\|_\diamond \le 2\varepsilon + \|\mathcal{P}_{\rho_i} - \mathcal{P}_{\rho_j}\|_\diamond \le 2\varepsilon + \|\rho_i - \rho_j\|_1.$$

□

Theorem 6.14 (Packing lower bound for physical programming). *Let $m \ge 2$, put $n = m^2 - 1$, and suppose $L \ge 2$ target unitary channels in $\mathbf{UChan}(H)$ are pairwise diamond-separated by at least Δ; for $L = 1$ the bound is trivial. Then*

$$\gamma_{m,d} \ge \max\left\{0, \frac{\Delta}{2} - \frac{2}{L^{1/n} - 1}\right\}.$$

The same right-hand side also lower-bounds the continuous-assignment quantity $\gamma^{\mathrm{cont}}_{m,d}$ defined below in Proposition 6.16; once that quantity is defined, the bound applies to it as well because continuous program encoders form a subclass of all program assignments.

The proof is given in Appendix A.4.

Corollary 6.15 (Explicit unitary-programming estimates). *The d^2 Weyl unitary channels are pairwise diamond-separated by two, so for $m \ge 2$,*

$$\gamma_{m,d} \ge \max\left\{0, 1 - \frac{2}{d^{2/(m^2-1)} - 1}\right\}.$$

For $m = d = 2$, $n = 3$, $L = 4$, this general bound gives $1 - 2/(4^{1/3} - 1) < 0$, hence only the trivial bound 0. A sharper qubit-specific packing argument using the Bloch sphere gives the following. For $m = 1$, $\gamma_{1,d} \ge 1$. For qubit targets and a qubit program,

$$\gamma_{2,2} \ge 1 - \frac{1}{2}\sqrt{\frac{8}{3}}.$$

Proof. Fix the standard shift and phase operators X and Z on $\mathbb{C}^d$. The d^2 operators $W_{a,b} = X^a Z^b$, $0 \le a, b < d$, indexed by $\mathbb{Z}_d^2$, are Hilbert–Schmidt orthogonal ($\mathrm{Tr}(W_{a,b}^\dagger W_{c,d}) = d\,\delta_{a,c}\delta_{b,d}$) and define d^2 distinct unitary channels up to global phase (global phase does not affect the channel). Hence maximally entangled inputs perfectly distinguish their unitary channels. The first bound follows from Theorem 6.14. A one-dimensional program gives one fixed channel, which is at distance at least one from one of two perfectly distinguishable targets.

For $m = d = 2$, write four program states as $\rho_i = (I + r_i \cdot \sigma)/2$, with Bloch vectors r_i in the unit ball. The Hermitian matrix $\rho_i - \rho_j = (r_i - r_j) \cdot \sigma/2$ has eigenvalues $\pm\|r_i - r_j\|_2/2$ because $(a \cdot \sigma)$ has eigenvalues $\pm\|a\|_2$, and hence $\|\rho_i - \rho_j\|_1 = \|r_i - r_j\|_2/2 + \|r_i - r_j\|_2/2 = \|r_i - r_j\|_2$. Therefore

$$\sum_{i<j} \|r_i - r_j\|_2^2 = 4\sum_i \|r_i\|_2^2 - \left\|\sum_i r_i\right\|_2^2 \le 16.$$

Among six pairs, one therefore has squared distance at most $8/3$. Proposition 6.13 gives the last bound; equality in the geometric estimate is attained by a regular tetrahedron. □

Model	Complete retained code	Decoder model	Applicable statement
Euclidean latent code	$\mathbb{R}^r$	unrestricted	fiber/profile bounds
Quantum-state alphabet	$\mathcal{S}(M)$, $r = m^2 - 1$	unrestricted	channel lower envelope
Classical–quantum code	blocks m_j, $r = \sum_j m_j^2 - 1$	unrestricted	channel envelope with the complete code dimension
Physical program	program state on M	fixed CPTP processor	$\gamma_{m,d} > 0$; continuous-assignment profile bound
Deterministic N-query network	complete final quantum–classical output	unrestricted or physical	bound uniform in finite N
Uniform postselection	normalised success state, $p \geq p_0$	as above	same source bound on the success domain
Finite sampled transcript	one sampled symbol from a c-element alphabet; its channel-dependent law $p(\cdot\|\Phi)$ lies in $\Sigma(T) \subseteq \mathbb{R}^{c-1}$ (transcript distribution), actual transcript sampled from that distribution	randomized transcript decoder (Borel Q_t)	collision lower bound α_{c-1}; IC $N^{-1/2}$ upper bound with growing alphabet

Table 2: Coding and reconstruction models. Classical transcripts and side registers count as part of the complete retained code.

Proposition 6.16 (Continuous physical-programming lower bound)**.** *Define*

$$\gamma_{m,d}^{\mathrm{cont}} = \inf_{\mathcal{P}} \inf_{f \in C(\mathbf{UChan}(H), \mathcal{S}(M))} \sup_{\mathcal{U} \in \mathbf{UChan}(H)} \|\mathcal{P}_{f(\mathcal{U})} - \mathcal{U}\|_\diamond.$$

Then

$$\gamma_{m,d}^{\mathrm{cont}} \geq \alpha_{m^2-1}(\mathbf{UChan}(H)).$$

If $m^2 - 1 \leq 2\lfloor \log_2 d \rfloor$, then

$$\gamma_{m,d}^{\mathrm{cont}} \geq 1.$$

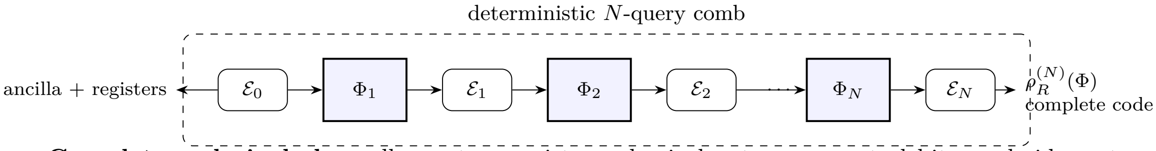
Proof. For fixed $\mathcal{P}$, the map $g(\rho) = \mathcal{P}_\rho$ is a decoder from $\mathcal{S}(M)$ to channels. Affine coordinates inject $\mathcal{S}(M)$ into $\mathbb{R}^{m^2-1}$, so the source–latent obstruction applied to the continuous encoder f gives the first inequality. Corollary 5.8 gives the second. Taking the two infima preserves both bounds. □

Remark 6.17 (Continuous versus arbitrary program assignments)*.* The quantity $\gamma_{m,d}$ permits a discontinuous assignment of program states, whereas $\gamma_{m,d}^{\mathrm{cont}}$ requires a continuous program encoder. Compactness proves positivity of the former but does not determine its value. Proposition 6.13 supplies the basic metric input for quantitative packing estimates.

Table 2 collects the decoder and retained-code assumptions used by the abstract and operational statements. Its first four rows summarize the models of this section; the final three preview the query-generated encoders treated in Section 7.

7 Adaptive query access

The previous section constrains a code once its encoder is known to be continuous. Here finite-query networks supply that continuity operationally. The hybrid estimate first controls deter-



Complete code includes: all quantum registers, classical outcomes, control bits, and side systems available to decoder. Discarding an outcome corresponds to averaging. Classical transcript $T \in \{1, \dots, c\}$ sampled from $p(\cdot|\Phi) \in \Sigma_{c-1}$ is part of $\rho_R^{(N)}$ when retained; its law is continuous in Φ (Proposition 7.1). Postselection with $p(\Phi) \geq p_0 > 0$ keeps continuity after normalisation (Proposition 7.5).

Figure 4: Deterministic finite-query adaptive network as a quantum comb. The N slots are filled with the unknown channel (or with slot-dependent channels Φ_j, as in the heterogeneous-sensitivity Corollary 7.2 stated below). All interleavings are fixed CPTP maps, possibly with coherent control and intermediate measurements. The final state $\rho_R^{(N)}$ is the complete retained code; any classical transcript, quantum memory M, or cq blocks (m_j) is a function of it. Lower bounds β_r depend only on $\operatorname{embdim}(f(\mathbf{Chan}))$ or $\operatorname{coind} F_2(Y)$, uniformly in N.

ministic retained states, after which the source bounds are applied to fixed final-code alphabets and uniformly successful postselection. Finite sampled transcripts are treated separately with both collision lower bounds and an explicit reconstruction protocol.

7.1 Deterministic networks and implementation sensitivity

Definition 7.1 (Deterministic finite-query network)**.** A deterministic N-query network is a causally ordered finite sequence of CPTP interleaving maps with the unknown channel inserted in N designated slots. Finite ancillas, coherent controls, intermediate measurements, and classically controlled operations are allowed. The complete final output includes every classical outcome, quantum register, and side system available to the decoder. If an outcome is discarded, the final code is the averaged state resulting from that discard. Postselection is excluded unless stated separately.

Figure 4 shows the complete retained code as the output of the comb. The following hybrid argument quantifies its sensitivity.

Proposition 7.1 (Hybrid Lipschitz estimate)**.** *Let $\rho_R^{(N)}(\Phi)$ be the complete final state produced by a deterministic N-query network. Then*

$$\|\rho_R^{(N)}(\Phi) - \rho_R^{(N)}(\Psi)\|_1 \leq N\|\Phi - \Psi\|_\diamond.$$

Proof. Order the slots causally. For $k = 0, \dots, N$, let $\rho^{(k)}$ be the final state when the first k slots use Φ and the remainder use Ψ. For $k = 1, \dots, N$, adjacent hybrids use identical channels and operations before slot k, so the joint state entering that slot is the same. Replacing one channel changes the trace norm by at most $\|\Phi - \Psi\|_\diamond$. This remains valid for every accumulated ancilla including arbitrary classical registers because the diamond norm is already stabilized at a reference dimension d_A [17], i.e., $\|\Phi - \Psi\|_\diamond = \sup_{\rho \in \mathcal{S}(A_0 \otimes A)} \|(id \otimes (\Phi - \Psi))(\rho)\|_1$ suffices for any extension. Subsequent deterministic CPTP maps contract trace norm. Therefore

$$\|\rho^{(k)} - \rho^{(k-1)}\|_1 \leq \|\Phi - \Psi\|_\diamond,$$

and summing over k proves the claim. □

Corollary 7.2 (Heterogeneous-slot implementation sensitivity)**.** *For two deterministic networks with the same interleaving maps and slot channels $\Phi_1, \dots, \Phi_N$ and $\widetilde{\Phi}_1, \dots, \widetilde{\Phi}_N$, respectively,*

$$\|\rho_R(\Phi_1, \dots, \Phi_N) - \rho_R(\widetilde{\Phi}_1, \dots, \widetilde{\Phi}_N)\|_1 \leq \sum_{j=1}^{N} \|\Phi_j - \widetilde{\Phi}_j\|_\diamond.$$

If the reconstruction decoder is L-Lipschitz from trace norm to diamond distance, the reconstructed maps differ by at most L times the same sum.

Proof. Form hybrids that replace the slot channels one at a time in causal order. The diamond norm bounds the change at the replaced slot, and all subsequent CPTP operations contract the trace norm. Summing the adjacent-hybrid bounds proves the first claim; the decoder estimate is its Lipschitz consequence. $\square$

7.2 Uniform query bounds and postselection

Let $\mathfrak{C}_N(M)$ denote the class of deterministic finite-dimensional N-query networks whose complete retained output is a state on an m-dimensional quantum system M. Separately, let $\mathfrak{C}_N^{\mathrm{cq}}(m_1, \ldots, m_c)$ denote networks whose complete retained output belongs to the classical–quantum alphabet with block dimensions $m_1, \ldots, m_c$.

Theorem 7.3 (Query-independent final-code bottleneck)**.** *Set $r = m^2 - 1$. For every $N \geq 1$, every $\mathcal{C}^{(N)} \in \mathfrak{C}_N(M)$, and every unrestricted decoder from $\mathcal{S}(M)$ to $\mathbf{TP}_{\mathrm{HP}}(A, B)$, the worst-case error is at least $\beta_r(A, B)$. Consequently,*

$$\inf_{N\geq 1} \inf_{\mathcal{C}^{(N)}\in\mathfrak{C}_N(M)} \inf_{\mathrm{Dec}_N:\mathcal{S}(M)\to\mathbf{TP}_{\mathrm{HP}}(A,B)} \sup_{\Phi\in\mathbf{Chan}(A,B)} \| \mathrm{Dec}_N(\rho_R^{(N)}(\Phi)) - \Phi\|_\diamond \geq \beta_r(A, B).$$

The same lower bound applies to the limit inferior of any sequence of such finite-query worst-case errors.

Proof. Proposition 7.1 gives continuity of every final encoder. Apply Theorem 6.2. The bound is independent of N, so it survives all displayed infima and every limit inferior. $\square$

Corollary 7.4 (Query-independent classical–quantum code bound)**.** *For a network in $\mathfrak{C}_N^{\mathrm{cq}}(m_1, \ldots, m_c)$, set*

$$r_{\mathrm{cq}} = \sum_{j=1}^{c} m_j^2 - 1.$$

Every unrestricted decoder from that complete retained alphabet has worst-case error at least $\beta_{r_{\mathrm{cq}}}(A, B)$, uniformly in every finite query count.

Proof. Proposition 6.5 gives the Euclidean embedding dimension of the retained alphabet. The final-code map is continuous by Proposition 7.1; apply the same source–latent argument as in Theorem 7.3. $\square$

Proposition 7.5 (Uniformly successful postselection)**.** *Let $\sigma_R^{(N)}(\Phi)$ be the subnormalized success-branch state of a finite-query instrument, and suppose $p(\Phi) = \operatorname{Tr} \sigma_R^{(N)}(\Phi) \geq p_0 > 0$ for all channels considered. Then the normalised conditional encoder is continuous and satisfies*

$$\left\| \frac{\sigma_R^{(N)}(\Phi)}{p(\Phi)} - \frac{\sigma_R^{(N)}(\Psi)}{p(\Psi)} \right\|_1 \leq \frac{2N}{p_0} \|\Phi - \Psi\|_\diamond.$$

If the success condition holds on all of $\mathbf{Chan}(A, B)$ and the complete normalised success output belongs to the same retained latent alphabet, the bounds $\beta_r(A, B)$ apply unchanged. If it holds only on a promised compact subset K_0, the same argument gives the source bound $\alpha_r(K_0)$; the full channel-body bounds apply only when K_0 contains the relevant antipodal sections.

Proof. A CP trace-nonincreasing map $\mathcal{E}$ is trace-norm contractive on Hermitian operators, which suffices for the states appearing below. Indeed, if $X = X_+ - X_-$ is the Jordan decomposition, then positivity and trace-nonincreasingness give

$$\|\mathcal{E}(X)\|_1 \leq \operatorname{Tr} \mathcal{E}(X_+) + \operatorname{Tr} \mathcal{E}(X_-) \leq \operatorname{Tr} X_+ + \operatorname{Tr} X_- = \|X\|_1.$$

After a slot is fixed in the hybrid argument, continuation along the chosen success branch is CP and trace-nonincreasing. Hence $\|\sigma(\Phi) - \sigma(\Psi)\|_1 \leq N\|\Phi - \Psi\|_\diamond$. For positive subnormalized states σ, τ with traces $p, q \geq p_0$,

$$\left\|\frac{\sigma}{p} - \frac{\tau}{q}\right\|_1 \leq \frac{\|\sigma - \tau\|_1}{p} + q\left|\frac{1}{p} - \frac{1}{q}\right|$$
$$\leq \frac{2}{p_0}\|\sigma - \tau\|_1.$$

Indeed, $q|1/p - 1/q| = |p - q|/p$ because $q|1/p - 1/q| = |q/p - 1| = |q - p|/p$, and $|p - q| = |\operatorname{Tr}\sigma - \operatorname{Tr}\tau| \leq \|\sigma - \tau\|_1$, so $q|1/p - 1/q| \leq \|\sigma - \tau\|_1/p$, and $p \geq p_0$. ☐

7.3 Finite sampled transcripts

A retained probability distribution is a deterministic point of a simplex, whereas an observed transcript is a random variable. The first result below bounds worst-source expected error from the dimension of the full transcript simplex; the subsequent informationally complete construction supplies a finite-sample upper bound.

Definition 7.2 (Sampled-transcript reconstruction error)**.** Let T be a finite transcript alphabet and let $p : K \to \Delta(T)$ be a continuous stochastic encoder. For each $t \in T$, let Q_t be a Borel probability distribution on decoder outputs in X (if expectations infinite, inequality trivial). Define

$$\mathcal{E}(x) = \sum_{t \in T} p(t|x) \int_X d(z, x)\, dQ_t(z).$$

Theorem 7.6 (Worst-case sampled-transcript obstruction)**.** *If $|T| = c < \infty$, then every continuous stochastic encoder and every possibly randomized transcript decoder satisfy*

$$\sup_{x \in K} \mathcal{E}(x) \geq \alpha_{c-1}(K).$$

Proof. If any displayed expected error is infinite or not well-defined, the conclusion is immediate (inequality interpreted trivially). Otherwise all integrals below are finite and well-defined. Identify the probability simplex $\Delta(T)$ with a subset of $\mathbb{R}^{c-1}$. For any $h : S^{c-1} \to K$, Borsuk–Ulam gives an antipodal pair $x_+ = h(u)$ and $x_- = h(-u)$ with the same transcript distribution. For every transcript t and decoder output z,

$$d(z, x_+) + d(z, x_-) \geq d(x_+, x_-).$$

Integrating first against Q_t and then against the common transcript distribution gives

$$\mathcal{E}(x_+) + \mathcal{E}(x_-) \geq d(x_+, x_-).$$

Thus one expected error is at least half the antipodal separation. Taking the supremum over h proves the result. ☐

Corollary 7.7 (Sampled channel transcript)**.** *Let a finite-query protocol output one sampled transcript from an alphabet of cardinality c, with transcript probabilities continuous in the unknown channel. Every possibly randomized reconstruction rule satisfies*

$$\sup_{\Phi \in \mathbf{Chan}(A,B)} \mathbb{E}_\Phi \|\widehat{\Phi}_T - \Phi\|_\diamond \geq \alpha_{c-1}(\mathbf{Chan}(A, B)) \geq \beta_{c-1}(A, B).$$

In particular, the lower bound is one whenever $c \leq d_B^2 - 1$.

Proof. Apply Theorem 7.6 with the channel body and diamond distance, followed by Theorem 5.16. ☐

Proposition 7.8 (One-query injective latent-state encoder)**.** *Write $\Sigma_D = \{p \in \mathbb{R}^{D+1} : p_i \geq 0,\ \sum_{i=0}^{D} p_i = 1\}$ for the D-dimensional probability simplex with $D+1$ vertices (i.e., probability distributions on $D+1$ symbols). For every sufficiently small $\eta > 0$, there is a one-query measurement (a POVM that is informationally complete for the channel affine hull, i.e. for the direction space $V_{A,B}$; cf. [22] for informationally complete measurements in general) whose outcome-probability map $\Phi \mapsto p(\Phi) \in \Sigma_{D_{A,B}}$, $p_i(\Phi) = \mathrm{Tr}(\widehat{J}_\Phi E_i)$, is a continuous affine injection from $\mathbf{Chan}(A, B)$ into the classical probability simplex. Equivalently, the averaged, pre-sampling classical output state $\sum_i p_i(\Phi)|i\rangle\langle i|$ lies in $\Sigma_{D_{A,B}}$ and its channel-to-state map is continuous, affine, and injective. A single run yields only a sampled outcome; the sampled version is treated in Theorem 7.9.*

Proof. Choose a Hilbert–Schmidt orthonormal basis $H_1, \dots, H_{D_{A,B}}$ of the direction space $V_{A,B}$. Every H_i is traceless. For $0 < \eta < 1/((D_{A,B}+1)\sum_i \|H_i\|_\infty)$, the operators

$$E_i = \frac{I_{A_0 \otimes B}}{D_{A,B}+1} + \eta H_i \quad (1 \leq i \leq D_{A,B}),$$

and

$$E_0 = \frac{I_{A_0 \otimes B}}{D_{A,B}+1} - \eta \sum_{i=1}^{D_{A,B}} H_i$$

are positive: each perturbing operator has operator norm at most $\eta \sum_i \|H_i\|_\infty < 1/(D_{A,B}+1)$, below the minimal eigenvalue $1/(D_{A,B}+1)$ of $I/(D_{A,B}+1)$. They sum to the identity and hence form a POVM.

Prepare $|\omega_A\rangle$, apply the unknown channel once, measure the resulting normalised Choi state, and retain the classical output state with probabilities

$$p_i(\Phi) = \mathrm{Tr}(\widehat{J}_\Phi E_i).$$

The map $\Phi \mapsto p(\Phi)$ is affine and continuous. If $p(\Phi) = p(\Psi)$ and $X = \widehat{J}_\Phi - \widehat{J}_\Psi$, then $X \in V_{A,B}$ and $\mathrm{Tr}\, X = 0$. For $1 \leq i \leq D_{A,B}$,

$$0 = \mathrm{Tr}(XE_i) = \eta\, \mathrm{Tr}(XH_i).$$

Since the H_i form a basis of $V_{A,B}$, this implies $X = 0$ and $\Phi = \Psi$. Thus the output-state encoder is injective. $\square$

Theorem 7.9 (Finite-sample channel reconstruction)**.** *Fix any admissible perturbation parameter $\eta > 0$ for the POVM constructed in Proposition 7.8 (any fixed η with $0 < \eta < 1/((D_{A,B}+1)\sum_i \|H_i\|_\infty)$); the resulting constant below depends on this choice and the statement is an existence statement. Repeat that one-query experiment independently N times. There is a channel-valued decoder from the sampled transcript such that*

$$\sup_{\Phi \in \mathbf{Chan}(A,B)} \mathbb{E}_\Phi \|\widehat{\Phi}_N - \Phi\|_\diamond \leq \min\left\{2,\ \frac{d_A\sqrt{d_A d_B}}{\eta\sqrt{N}}\right\}.$$

The ordered raw transcript alphabet has cardinality $(D_{A,B}+1)^N$. Because the decoder below uses only outcome counts, it may instead retain the count vector, whose alphabet has cardinality $\binom{N+D_{A,B}}{D_{A,B}}$.

Proof. Write

$$\widehat{J}_\Phi - \widehat{J}_{\Omega_{A,B}} = \sum_{i=1}^{D_{A,B}} x_i(\Phi) H_i.$$

Since $\widehat{J}_{\Omega_{A,B}}$ is scalar and the H_i are traceless,

$$p_i(\Phi) = \frac{1}{D_{A,B}+1} + \eta x_i(\Phi) \quad (1 \le i \le D_{A,B}).$$

Let q_i be the empirical frequency of outcome i and set

$$y_i = \frac{q_i - 1/(D_{A,B}+1)}{\eta}.$$

Let $c(\Phi) = (x_i(\Phi))_{i=1}^{D_{A,B}}$ and let $C = c(\mathbf{Chan}(A,B))$. Project y orthogonally onto C, and decode the projected point to a channel $\widehat{\Phi}_N$. Euclidean projection onto a closed convex set is nonexpansive, so

$$\|c(\widehat{\Phi}_N) - c(\Phi)\|_2 \le \frac{1}{\eta}\|q_{1:D_{A,B}} - p_{1:D_{A,B}}(\Phi)\|_2.$$

For $\Delta = \widehat{\Phi}_N - \Phi$, Lemma 3.2 gives $\|\Delta\|_\diamond \le \|\operatorname{Tr}_B |J_\Delta|\|_\infty$, and $\|\operatorname{Tr}_B |J_\Delta|\|_\infty \le \|\operatorname{Tr}_B |J_\Delta|\|_1 \le \|J_\Delta\|_1$ because operator norm $\le$ trace norm and partial trace is trace-norm contractive. Cauchy–Schwarz then gives

$$\begin{aligned}\|\Delta\|_\diamond &\le \|\operatorname{Tr}_B |J_\Delta|\|_\infty \le \|J_\Delta\|_1 \\ &\le d_A\sqrt{d_A d_B}\,\|\widehat{J}_\Delta\|_2 = d_A\sqrt{d_A d_B}\,\|c(\widehat{\Phi}_N) - c(\Phi)\|_2.\end{aligned}$$

The multinomial covariance identity gives

$$\mathbb{E}_\Phi\|q_{1:D_{A,B}} - p_{1:D_{A,B}}(\Phi)\|_2^2 = \frac{1}{N}\left(\sum_{i=1}^{D_{A,B}} p_i(\Phi) - \sum_{i=1}^{D_{A,B}} p_i(\Phi)^2\right) \le \frac{1}{N}.$$

Jensen's inequality proves the decay bound, and the minimum with 2 is admissible because the channel body has diamond diameter 2 for $d_B > 1$ (Section 5), so the displayed estimate is non-vacuous for every N; the decay is the asymptotic content. The decoder is defined as above on the encoder image (count vectors) and set to the fixed channel $\Omega_{A,B}$ outside that image; no regularity is required. Concentration follows from Hoeffding's inequality, and sample-optimal state tomography bounds are given in [23]. □

Remark 7.10 (Latent states and sampled outcomes). Proposition 7.8 concerns the averaged classical state and the unrestricted decoder model. One execution supplies one sampled outcome, not the exact probability vector. Theorem 7.9 gives the corresponding operational finite-sample statement; its growing transcript alphabet is consistent with Theorem 7.6.

Remark 7.11 (Sampled transcripts and retained information). This is a worst-channel expected-error statement, not a Bayes-average bound for an arbitrary nonatomic prior. If a length-N transcript uses a c-symbol alphabet, the complete transcript alphabet can have up to c^N values; if only a statistic with L values is retained, $L-1$ is the relevant simplex dimension. The decoder receives only that complete retained code and has no subsequent access to the unknown channel. Every classical transcript, quantum register, reference system, or side register available to it is part of the code. A sampled transcript with expected-error performance is a probabilistic model and is not identified with the deterministic average output state.

Remark 7.12 (Continuity is essential). Without encoder regularity, a set-theoretic injection of the channel body into another continuum, followed by its inverse on the image, gives exact reconstruction. The obstruction is therefore topological rather than cardinality-based: continuity of the encoder is indispensable, while the decoder may remain unrestricted.

Remark 7.13 (Continuity versus the Lipschitz constant). Continuity alone activates the exact antipodal collision and the topological lower bound. The factor N quantifies implementation sensitivity but does not weaken the lower bound as N grows. It becomes relevant in noisy-code or Lipschitz-decoder variants such as Theorem 2.12.

8 Conclusion and open problems

Encoder-continuous reconstruction admits an exact fiber-radius formulation. The antipodal profile quantifies source geometry and the deleted-product coindex quantifies latent collision topology. The replacement family, the observable sphere, and their join provide the maximal lower bound 1 through index $L_{A,B}$; the Clifford construction extends perfect separation to unitary channels. Across the remaining subcritical dimensions the globally optimal centred ball and the full-direction section give the certified lower bound $\gamma_{A,B} = \max\{\rho_{\text{dep}}(A, B), 1/d_A\}$ up to the affine dimension $D_{A,B} = d_A^2(d_B^2 - 1)$, which is the exact threshold for zero-error continuous encoding with unrestricted decoder. The n-outcome instrument versions of these results appear in Ref. [11]. The depolarizing channel is an optimal constant decoder with exact covering radius $2 - \rho_{\text{dep}}(A, B)$. The lower envelope is complemented by constructive upper envelopes from convex projection and balanced input/output pinching, as illustrated in Figure 3 and Figure 2, bracketing the unknown positive widths; at $r = 0$ the two envelopes meet at the exact constant-code width. For abstract quantum-state codes the bounds imply $m \geq d_B$ is necessary for error below one and $m^2 - 1 \geq D_{A,B}$ is necessary and sufficient for exact encoding.

The Euclidean zero-error threshold is not a universal-programming threshold. Every deterministic finite-query adaptive network, equivalently a finite quantum comb, induces a continuous final code containing all classical and quantum side information (Figure 4); the same dimensional bottlenecks hold uniformly in the query count, for uniformly successful postselection, and for finite sampled transcripts. A one-query informationally complete measurement yields a continuous injective encoding at the affine threshold, and repeated sampling yields a channel-valued estimator with expected error $O(N^{-1/2})$. These results isolate final representational dimension as the obstruction, distinct from tomography sample complexity and from fixed-processor programming limits.

Several concrete problems remain:

1. determine the exact positive widths $\widetilde{\delta}_r^{\diamond}(A, B)$ and $\delta_r^{\diamond}(A, B)$ for $0 < r < D_{A,B}$ (for $d_A = 1$, the channel-level instance of the flat-widths problem of Ref. [11] is settled in the negative direction: the flat value $\delta_r^{\diamond} = 1$ fails at the bottom index for $d_B \geq 5$ and, more generally, at every $r \leq (d_B - 3)/2$ within $1 \leq r \leq d_B^2 - 2$, by the coordinate-flag median matching and topological Tverberg bounds of the companion — at $d_A = 1$ the diamond norm on channel differences is the trace norm on output states, so the channel widths coincide with the one-outcome instrument widths; and the companion's equal-value-basis theorem closes those two cases as well: the flat value fails for every $d_B \geq 3$ at the bottom index, with $\delta_1^{\diamond} \geq 4/3$ — the qutrit channel state space exactly at 4/3 and the ququart in [4/3, 3/2]);

2. determine the exact perfect-separation indices $\pi(A, B)$ and $\pi_{\text{U}}(d)$ of Corollary 5.18 within the proved lower and upper ranges;

3. determine exact deleted-product coindices of structured nonconvex latent spaces beyond the patch and embedding bounds proved here;

4. determine whether the optimal Chebyshev centre $\Omega_{A,B}$ is unique;

5. sharpen the explicit physical-programming packing bounds and develop Bayes-risk or infinite-transcript formulations beyond the finite-alphabet worst-case theorem.

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A Technical proofs

The appendix provides the complete matrix and volume proofs for the full-body radius and covering results, the fixed-marginal and programming estimates, and the structured partition and Clifford constructions.

A.1 Exact all-dimensional centred in-radius

Proof of Theorem 4.1. Let $s = \min\{d_A, d_B\}$. First consider an arbitrary nonzero Hermiticity-preserving, trace-annihilating map Δ, put $X = \widehat{J}_\Delta$, and let

$$\lambda = \lambda_{\max}(-X).$$

Choose a unit vector $|v\rangle$ satisfying $X|v\rangle = -\lambda|v\rangle$. Let $S \subseteq A_0$ be the support of $\operatorname{Tr}_B |v\rangle\langle v|$ and set $b = \dim S$. The Schmidt rank of $|v\rangle$ is b, so $b \le s$.

Choose an orthonormal basis of S and use the basis-conjugate subspace of A under the fixed identification $A_0 \simeq A$. On these two subspaces let

$$|\omega_S\rangle = \frac{1}{\sqrt{b}} \sum_{j=1}^{b} |j\rangle_{A_0} |j\rangle_A.$$

If P_S is the projection onto S, then the normalised-Choi convention gives

$$Y := (\mathrm{id}_{A_0} \otimes \Delta)(|\omega_S\rangle\langle\omega_S|) = \frac{d_A}{b}(P_S \otimes I_B) X (P_S \otimes I_B).$$

The operator Y is Hermitian and traceless because Δ is trace-annihilating. Since $|v\rangle \in S \otimes B$,

$$\langle v|Y|v\rangle = -\frac{d_A}{b}\lambda.$$

For a traceless Hermitian operator $Y = Y_+ - Y_-$, $\operatorname{Tr} Y_+ = \operatorname{Tr} Y_- = \frac{1}{2}\|Y\|_1$ and for any unit vector $|v\rangle$, $\langle v|Y|v\rangle \ge -\|Y_-\|_\infty \ge -\operatorname{Tr} Y_- = -\frac{1}{2}\|Y\|_1$. Hence

$$\|Y\|_1 \ge -2\langle v|Y|v\rangle = \frac{2d_A}{b}\lambda \ge \frac{2d_A}{s}\lambda.$$

Because $|\omega_S\rangle\langle\omega_S|$ is an admissible input in the diamond-norm optimization,

$$\|\Delta\|_\diamond \ge \|Y\|_1.$$

Therefore

$$\lambda_{\max}(-\widehat{J}_\Delta) \le \frac{s}{2d_A}\|\Delta\|_\diamond,$$

and consequently $\Gamma_{A,B} \le s/(2d_A)$.

For the matching witness, choose an s-dimensional subspace $S_A \subseteq A$ and two isometries $V_0, V_1 : S_A \to B$ satisfying $\operatorname{Tr}(V_0^\dagger V_1) = 0$. If $s = 1$, the standing assumption $d_B > 1$ allows two orthogonal output unit vectors. If $s \ge 2$, take $V_1 = V_0 U$ for a traceless unitary U on S_A. Let

$$|v_j\rangle = (I \otimes V_j)|\omega_{S_A}\rangle, \qquad P_j = |v_j\rangle\langle v_j|, \quad j = 0, 1.$$

Then $P_0 P_1 = 0$ and

$$\operatorname{Tr}_B P_0 = \operatorname{Tr}_B P_1 = \frac{P_{S_A}}{s}.$$

Define completely positive trace-nonincreasing maps

$$\mathcal{E}_j(Z) = V_j P_{S_A} Z P_{S_A} V_j^\dagger.$$

For every input operator Z, $\operatorname{Tr}\mathcal{E}_j(Z) = \operatorname{Tr}(P_{S_A}Z)$; hence every scalar multiple of $\mathcal{E}_1 - \mathcal{E}_0$ is trace-annihilating. Each $\mathcal{E}_j$ has diamond norm at most one. A maximally entangled input on S_A produces the orthogonal pure outputs P_0 and P_1, so

$$\|\mathcal{E}_1 - \mathcal{E}_0\|_\diamond = 2.$$

Moreover,

$$\widehat{J}_{\mathcal{E}_j} = \frac{s}{d_A} P_j.$$

Thus the map

$$\widetilde{\Delta} = \frac{1}{2}(\mathcal{E}_1 - \mathcal{E}_0)$$

has diamond norm one and normalised Choi operator

$$\widehat{J}_{\widetilde{\Delta}} = \frac{s}{2d_A}(P_1 - P_0).$$

It follows that

$$\lambda_{\max}(-\widehat{J}_{\widetilde{\Delta}}) = \frac{s}{2d_A},$$

which proves $\Gamma_{A,B} = s/(2d_A)$. Proposition 4.3 then yields

$$\rho_\diamond(\Omega_{A,B}) = \frac{1}{d_A d_B \Gamma_{A,B}} = \frac{2}{d_B s}.$$

□

A.2 Exact Chebyshev covering radius

Proof of Theorem 5.20. Let $s = \min\{d_A, d_B\}$. Recall the diamond norm is already achieved with a reference system of dimension d_A, so we may assume $r = \operatorname{rank}\rho_R \le d_A$. A bipartite state σ_{RB} with marginal ρ_R and $r = \operatorname{rank}\rho_R$ satisfies

$$\sigma_{RB} \le k\,\rho_R \otimes I_B, \qquad k = \min\{r, d_B\}.$$

To prove this, restrict R to the support of ρ_R and set, with $\rho_R^{-1/2}$ denoting the Moore-Penrose inverse on its support,

$$Z = (\rho_R^{-1/2} \otimes I_B)\sigma_{RB}(\rho_R^{-1/2} \otimes I_B).$$

If $\sigma_{RB} = 0$ then $\rho_R = 0$ and the claim is immediate, so assume $\sigma_{RB} \ne 0$. Then $Z \ge 0$ and $\operatorname{Tr}_B Z = I_R$. With the column-vectorization convention of Section 3, $|K\rangle\rangle = \sum_{a,b} K_{ba}|a\rangle_R \otimes |b\rangle_B$, so that $\operatorname{Tr}_B |K\rangle\rangle\langle\langle K| = K^\dagger K$ and $\langle\langle V|K\rangle\rangle = \operatorname{Tr}(V^\dagger K)$, write $Z = \sum_\alpha |K_\alpha\rangle\rangle\langle\langle K_\alpha|$, where the condition $\operatorname{Tr}_B Z = I_R$ is equivalent to $\sum_\alpha K_\alpha^\dagger K_\alpha = I_R$. Let a unit vector $|v\rangle$ of Schmidt rank $b \le k$ correspond to a matrix $V : R \to B$ with $\operatorname{Tr}(V^\dagger V) = 1$. Then $\operatorname{rank} V = b$. For each α, the matrix $K_\alpha V^\dagger$ has rank at most b. The elementary rank–trace inequality $|\operatorname{Tr} M|^2 \le \operatorname{rank}(M)\operatorname{Tr}(MM^\dagger)$, obtained by Cauchy–Schwarz on the support of M (write M via SVD $M = \sum_{i=1}^{\operatorname{rank} M} s_i |u_i\rangle\langle w_i|$, then $|TrM|^2 \le \operatorname{rank} M \sum s_i^2 |\langle w_i|u_i\rangle|^2 \le \operatorname{rank} M\, Tr(MM^\dagger)$), therefore gives

$$|\operatorname{Tr}(V^\dagger K_\alpha)|^2 = |\operatorname{Tr}(K_\alpha V^\dagger)|^2 \le b\ \operatorname{Tr}(V K_\alpha^\dagger K_\alpha V^\dagger).$$

Consequently,

$$\begin{aligned}\langle v|Z|v\rangle &= \sum_\alpha |\langle v|K_\alpha\rangle\rangle|^2 \\ &\le b\ \operatorname{Tr}\left(V\left(\sum_\alpha K_\alpha^\dagger K_\alpha\right)V^\dagger\right) = b \le k.\end{aligned}$$

Thus $Z \leq kI$, which proves the domination inequality. Since $r \leq d_A$, it follows that

$$\sigma_{RB} \leq d_B s \left(\rho_R \otimes \frac{I_B}{d_B} \right).$$

Both $\sigma_{RB} = (\mathrm{id}_R \otimes \Phi)(\xi_{RA})$ and $\tau = \xi_R \otimes I_B/d_B$ have trace one since $(\mathrm{id}_R \otimes \Phi)$ is itself a channel. If states ρ and τ satisfy $\rho \leq K\tau$, let P be the positive spectral projection of $\rho - \tau$ and put $a = \mathrm{Tr}(P\rho)$ and $b = \mathrm{Tr}(P\tau)$. Then $a \leq Kb$ because $P\rho P \leq KP\tau P$, and $a \leq 1$, so

$$\frac{1}{2}\|\rho - \tau\|_1 = a - b \leq a\left(1 - \frac{1}{K}\right) \leq 1 - \frac{1}{K}.$$

Applying this with $K = d_B s$ to $\sigma_{RB} = (\mathrm{id}_R \otimes \Phi)(\xi_{RA})$ and $\tau = \xi_R \otimes I_B/d_B$ yields

$$\|\Phi - \Omega_{A,B}\|_\diamond \leq 2\left(1 - \frac{1}{d_B s}\right).$$

For the reverse inequality, choose an s-dimensional subspace $S \subseteq A$, an isometry $V : S \to B$, and a state τ on B. With P_S the subspace projection, define the channel

$$\Phi(X) = VP_S X P_S V^\dagger + \mathrm{Tr}[(I_A - P_S)X]\,\tau.$$

A maximally entangled input of Schmidt rank s supported on S produces a pure maximally entangled output, whereas the depolarizing channel produces $I_{R_sB}/(sd_B)$. Their trace norm is

$$2\left(1 - \frac{1}{sd_B}\right).$$

Theorem 5.19 then gives the asserted constant-code equalities. □

A.3 One-sided twirling with a fixed marginal

Proof of Proposition 5.22. For an operator Y on $B_1 \otimes B_2$, one-sided Haar twirling gives

$$\int_{\mathrm{U}(B_2)} (I_{B_1} \otimes V)Y(I_{B_1} \otimes V^\dagger)\,dV = \mathrm{Tr}_{B_2}(Y) \otimes \frac{I_{B_2}}{d_{B_2}}.$$

Indeed, expand Y in matrix units on B_1 and apply the scalar Haar-twirl identity to every B_2 block.

Let $R_K(\Psi) = \sup_{\Phi \in K} \|\Phi - \Psi\|_\diamond$ on the affine class of centres with marginal Λ. Invariance of K makes R_K invariant under post-composition by the conjugation channel $\mathrm{id}_{B_1} \otimes \mathcal{C}_V$, and R_K is convex. The one-sided twirl of every admissible centre is

$$X \longmapsto \Lambda(X) \otimes \frac{I_{B_2}}{d_{B_2}} = \Phi_0(X).$$

Jensen's inequality therefore gives $R_K(\Phi_0) \leq R_K(\Psi)$ for every admissible centre. □

A.4 Physical-programming packing bound

Proof of Theorem 6.14. Fix an arbitrary processor $\mathcal{P}$ and put

$$e(\mathcal{P}) = \sup_{\mathcal{U} \in \mathbf{UChan}(H)} \min_{\rho \in \mathcal{S}(M)} \|\mathcal{P}_\rho - \mathcal{U}\|_\diamond.$$

Choose target channels $\Phi_1, \ldots, \Phi_L$ with pairwise diamond distance at least Δ. Compactness of the program-state space gives, for every target, a minimizing program state ρ_i (minimum attained because $\mathcal{S}(M)$ is compact and $\rho \mapsto \|\mathcal{P}_\rho - \mathcal{U}\|_\diamond$ is continuous, indeed $\|\mathcal{P}_\rho - \mathcal{P}_\sigma\|_\diamond \leq \|\rho - \sigma\|_1$), and its error is at most $e(\mathcal{P})$. Proposition 6.13 therefore gives

$$\|\rho_i - \rho_j\|_1 \geq \eta := \Delta - 2e(\mathcal{P}).$$

If $\eta = \Delta - 2e(\mathcal{P}) \leq 0$, then $e(\mathcal{P}) \geq \Delta/2$ and the asserted lower bound is immediate (with positive-part convention $\max\{0, \cdot\}$). Assume $\eta > 0$ and translate the affine space of trace-one Hermitian matrices by ρ_1. It is a real normed space of dimension $n = m^2 - 1$. All translated program states lie in the trace-norm ball of radius two, and the open balls of radius $\eta/2$ about them are disjoint. These balls lie in the ball of radius $2 + \eta/2$ about the origin. Fix any Lebesgue measure on this n-dimensional real vector space. Its norm balls scale in volume as the nth power of the radius up to a constant depending on the norm shape which cancels in the ratio, i.e., $\mathrm{Vol}(B(0,r)) = r^n \mathrm{Vol}(B(0,1))$, so

$$L\left(\frac{\eta}{2}\right)^n \leq \left(2 + \frac{\eta}{2}\right)^n.$$

Hence

$$\eta \leq \frac{4}{L^{1/n} - 1}.$$

Substituting $\eta = \Delta - 2e(\mathcal{P})$ and rearranging gives

$$e(\mathcal{P}) \geq \frac{\Delta}{2} - \frac{2}{L^{1/n} - 1}.$$

The processor was arbitrary, so taking its infimum proves the claimed bound for $\gamma_{m,d}$. Continuous program encoders form a subclass of all program assignments, so the same estimate applies to them. □

A.5 Structured partition sections

Proposition A.1 (Orthogonal-block join)**.** *Let $B = \bigoplus_{i=1}^{\ell} B_i \oplus B_\perp$. Suppose $h_i : S^{r_i} \to \mathcal{S}(B_i)$ are continuous and*

$$\|h_i(u) - h_i(-u)\|_1 \geq 2\rho_i.$$

Then there is a continuous

$$H : S^{\sum_i r_i + \ell - 1} \to \mathcal{S}(B)$$

whose antipodal trace norm is at least $2 \min_i \rho_i$.

Proof. Regard the domain sphere as the unit sphere in $\bigoplus_i \mathbb{R}^{r_i+1}$. For $y = (y_1, \ldots, y_\ell)$, put $\lambda_i = \|y_i\|_2^2$ and, when $y_i \neq 0$, $u_i = y_i/\|y_i\|_2$. Define the ith block of $H(y)$ to be $\lambda_i h_i(u_i)$, and set it to zero when $y_i = 0$. The latter extension is continuous because the block trace norm is λ_i. Since $\sum_i \lambda_i = 1$, $H(y)$ is a state. Orthogonality of the blocks gives

$$\begin{aligned}\|H(y) - H(-y)\|_1 &= \sum_i \lambda_i \|h_i(u_i) - h_i(-u_i)\|_1 \\ &\geq 2 \min_i \rho_i.\end{aligned}$$

□

Theorem A.2 (Partitioned replacement sections)**.** *Let $b_1, \ldots, b_\ell \geq 2$ satisfy $\sum_i b_i \leq d_B$, and put*

$$R_{\mathrm{J}} = \sum_i (b_i^2 - 1) - 1,$$

the block-join index. Then

$$\alpha_r(\mathbf{Chan}(A, B)) = 1, \qquad 0 \leq r \leq R_{\mathrm{J}}.$$

Proof. On a block of dimension $b_i \geq 2$, the Jordan construction in the proof of Theorem 5.3, restricted to $\mathrm{Herm}_0(B_i) \subseteq \mathrm{Herm}_0(B)$, gives a continuous map $h_i : S^{b_i^2-2} \to \mathcal{S}(B_i)$ whose antipodes have orthogonal supports, hence trace norm 2. Proposition A.1 produces a sphere of dimension

$$\sum_i (b_i^2 - 2) + \ell - 1 = \sum_i (b_i^2 - 1) - 1 = R_{\mathrm{J}}$$

with antipodal trace norm 2. Replacement channels preserve this distance by Lemma 5.1, so the antipodal separation in diamond distance is 2; dividing by two gives the profile value one on this sphere. Restriction to coordinate equators gives every smaller r, and the diameter bound supplies the reverse inequality. □

Corollary A.3 (Block-size envelope)**.** *Fix a block-size cap $2 \leq c \leq d_B$ and write $d_B = mc + t$, $0 \leq t < c$. Set*

$$R_{\mathrm{J}}(c) = \begin{cases} m(c^2 - 1) - 1, & t = 0 \text{ or } t = 1, \\ m(c^2 - 1) + t^2 - 2, & 2 \leq t < c. \end{cases}$$

Then

$$\alpha_r(\mathbf{Chan}(A, B)) = 1, \qquad 0 \leq r \leq R_{\mathrm{J}}(c).$$

In particular, for $c = 2$ (two-dimensional blocks),

$$\alpha_r(\mathbf{Chan}(A, B)) = 1 \quad \text{for} \quad 0 \leq r \leq 3 \left\lfloor \frac{d_B}{2} \right\rfloor - 1.$$

Proof. If $t = 0$ or $t = 1$, use m blocks of size c and leave the possible one-dimensional remainder unused. Theorem A.2 then gives $R_{\mathrm{J}}(c) = m(c^2 - 1) - 1$. If $2 \leq t < c$, add one block of size t, giving $R_{\mathrm{J}}(c) = m(c^2 - 1) + (t^2 - 1) - 1$. For $c = 2$ this is $3\lfloor d_B/2 \rfloor - 1$ for both parities of d_B. Theorem A.2 gives the value one on this range, and the diameter bound supplies the reverse inequality. □

Proposition A.4 (Optimization within a block-size constraint)**.** *Fix a block-size cap $2 \leq c \leq d_B$. Among all partitions with $2 \leq b_i \leq c$ and $\sum_i b_i \leq d_B$, the largest value of $\sum_i (b_i^2 - 1) - 1$ is $R_{\mathrm{J}}(c)$ from Corollary A.3.*

Proof. It is enough to maximise $F = \sum_i (b_i^2 - 1)$, because the final subtraction by one is fixed. A maximizing partition leaves at most one output dimension unused: if at least two dimensions were unused, one could add a block of size between two and c, strictly increasing F.

We next show that a maximizer has at most one block smaller than c. Suppose two such blocks have sizes $2 \leq a \leq b < c$. If $a + b \leq c$, replace them by one block of size $a + b$; the change in F is

$$(a + b)^2 - 1 - (a^2 - 1) - (b^2 - 1) = 2ab + 1 > 0.$$

If $a + b = c + 1$, replace them by one c-block and leave one dimension unused; the change is

$$c^2 - 1 - (a^2 - 1) - (b^2 - 1) = 2(a - 1)(b - 1) > 0.$$

If $a + b \geq c + 2$, replace them by blocks of sizes c and $a + b - c$, the latter lying between two and $c - 2$. The change is

$$(c^2 - 1) + ((a + b - c)^2 - 1) - (a^2 - 1) - (b^2 - 1) = 2(c - a)(c - b) > 0.$$

Each operation is admissible, preserves the used dimension except in the middle case, and strictly increases F. Hence no maximizing partition can contain two subfull blocks.

Hence let $u \in \{0, 1\}$ be the unused dimension of a maximizer. If there is a subfull block of size b, then $u = 0$: otherwise enlarging that block from b to $b + 1 \leq c$ would increase F by $2b + 1$ and keep $\sum b_i \leq d_B$. Thus the maximizer is either k blocks of size c with $u \in \{0, 1\}$ unused, or

k blocks of size c together with one block of size $b \in \{2, \dots, c-1\}$ and no unused dimension. Writing $d_B = mc + t$, $0 \le t < c$, the first form occurs exactly for $t = 0, 1$ and has $k = m$; the second occurs for $2 \le t < c$ and has $k = m$, $b = t$. In particular, if fewer than m full c-blocks remain, the remaining unused dimensions are $< c$ but at least 2 would allow adding or enlarging a block to increase F, so the optimizer must have m full c-blocks (with the remainder t handled as above). Therefore

$$\max F = \begin{cases} m(c^2-1), & t = 0 \text{ or } t = 1, \\ m(c^2-1) + (t^2-1), & 2 \le t < c. \end{cases}$$

Subtracting one gives $R_{\mathrm{J}}(c)$. □

A.6 Clifford replacement and unitary sections

Proof of Proposition 5.6 and Theorem 5.7. Identify C with $(\mathbb{C}^2)^{\otimes q}$ and define

$$\begin{aligned} \Gamma_{2j-1} &= \sigma_z^{\otimes(j-1)} \otimes \sigma_x \otimes I^{\otimes(q-j)}, \\ \Gamma_{2j} &= \sigma_z^{\otimes(j-1)} \otimes \sigma_y \otimes I^{\otimes(q-j)}, \qquad 1 \le j \le q, \\ \Gamma_{2q+1} &= \sigma_z^{\otimes q}. \end{aligned}$$

Each generator is a nonidentity Pauli string, hence is traceless and Hermitian, and its square is I_C. The two generators with the same active tensor position anticommute because $\sigma_x\sigma_y = -\sigma_y\sigma_x$. If one active position precedes the other, its σ_x or σ_y meets a σ_z at that position and all remaining tensor factors contribute no further sign. The last generator anticommutes with each preceding one in the same way. Thus

$$\Gamma_j\Gamma_k + \Gamma_k\Gamma_j = 2\delta_{jk} I_C.$$

Distinct Pauli strings are orthogonal in the Hilbert–Schmidt inner product, so the generators are linearly independent. For $u = (u_1, \dots, u_{2q+1}) \in S^{2q}$ (so $\sum_{j=1}^{2q+1} u_j^2 = 1$), let $H_u = \sum_{j=1}^{2q+1} u_j\Gamma_j$ summed over all $2q+1$ generators. The anticommutation relations give $H_u^2 = I_C$, while $\operatorname{Tr} H_u = 0$. Hence its eigenvalues are ± 1 with equal total multiplicity. Extended by zero to B, it satisfies $H_u^2 = P_C$. The states

$$\sigma_u = \frac{P_C + H_u}{p}$$

are positive, have trace one, and satisfy $\sigma_u\sigma_{-u} = (P_C - H_u^2)/p^2 = 0$. Linear independence of the generators also shows that $\sigma_u = \sigma_v$ implies $u = v$. Their antipodes have orthogonal supports, so their trace norm, and hence the diamond distance of the associated replacement channels, is 2.

For the unitary construction on a square system H, set

$$U_u^{(C)} = \frac{I_C + iH_u}{\sqrt{2}}, \qquad \widetilde{U}_u = U_u^{(C)} \oplus I_{C^\perp}.$$

Then $U_{-u}^{(C)} = (U_u^{(C)})^\dagger$. A maximally entangled input supported on the p-dimensional subspace C gives output vectors with inner product

$$\frac{1}{p}\operatorname{Tr}(((U_u^{(C)})^\dagger)^2) = -\frac{i}{p}\operatorname{Tr} H_u = 0.$$

Thus the antipodal channels generated by $\widetilde{U}_u$ and $\widetilde{U}_{-u}$ are perfectly distinguishable. □

Data Availability

No data were generated or analysed in this study.

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